Climate-Conditioned Comparative Advantage, Persistence, and U.S. Corn–Soy Land Allocation

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Abstract

Climate change can alter agricultural comparative advantage, but land allocation may adjust slowly when production patterns are persistent. We study county-level corn–soy allocation in the U.S. Corn Belt. Using rain-fed potential-yield simulations from FAO Global Agro-Ecological Zones, we construct a Relative Input Premium Advantage ($RIPA$) comparing potential-yield gains from higher-input management for corn and soybeans. $RIPA$ is associated mainly with persistent cross-county differences in allocation, while year-to-year acreage changes are more closely related to expected relative revenues. We estimate a discrete-choice land-allocation model incorporating expected portfolio revenue, $RIPA$, revenue risk, and prior allocation. The estimates show strong state dependence that remains under alternative spatial controls. Counterfactual acreage responses increase substantially when the estimated lag-distance penalties are removed. Historical GAEZ $RIPA$ shifts between the reference periods ending in 1990 and 2010 are geographically heterogeneous and imply modest aggregate reallocation under the estimated persistence penalties.

Keywords: Climate adaptation, comparative advantage, land allocation, state dependence, agro-climatic potential.

JEL Classification: Q12, Q15, Q54, C35.

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1 Introduction

Climate change can alter agricultural comparative advantage by changing the relative productivity of crops across locations. Adaptation can therefore occur through land reallocation toward crops that become relatively more attractive as production conditions change. Such adjustment need not be immediate. Crop rotations, machinery, management experience, input procurement, marketing relationships, and other persistent features of local production can slow changes in land use. Changes in comparative advantage may therefore translate only gradually into observed acreage adjustment.

A large literature documents the effects of temperature and precipitation on agricultural productivity (e.g., Mendelsohn and Nordhaus, 1994; Schlenker and Roberts, 2009) and examines adaptation to changing climatic conditions (e.g., Burke and Emerick, 2016; Cui, 2020; Taylor, 2026). Crop choice and the spatial allocation of production are important adaptation margins because climate change can alter the relative productivity of crops and locations (Costinot et al., 2016; Cui, 2020). Historical evidence also shows persistent agricultural responses to changes in local production conditions (Hornbeck, 2012), while equilibrium estimates of climate impacts depend on adjustment across crops, locations, and markets (Costinot et al., 2016; Gouel, 2025). When production patterns are persistent, short-run acreage responses can be small even when relative production incentives change substantially. We study how climate-conditioned production opportunities translate into corn–soy land allocation when expected returns, revenue risk, and persistence shape acreage decisions.

The U.S. Corn Belt provides a useful setting for this question. Corn and soybeans compete for much of the same agricultural land but respond differently to local climatic and biophysical conditions. Their geographic footprint has expanded northward and westward in recent decades, particularly into the Dakotas and other areas along the northern and western margins of the traditional Corn Belt (Du et al., 2014; Green et al., 2018; O'Brien et al., 2020; Wright and Wimberly, 2013). Both crops have expanded, but their relative growth differs across locations. The historical changes in relative corn–soy production opportunity documented below are also spatially heterogeneous, with larger corn-favoring shifts in parts of the northern and western production region. This correspondence is descriptive rather than causal. Our question is how changes in relative agro-climatic production opportunities translate into corn–soy allocation and how strongly existing production patterns dampen that response.

Technology and agro-climatic conditions jointly shape crop productivity. The gains from seeds, machinery, fertilizer, and management depend in part on local climatic conditions. For crop allocation, the relevant margin is relative: favorable conditions for corn need not favor corn acreage if they also favor soybeans.

We measure this relative production margin using potential-yield data from the FAO Global Agro-Ecological Zones (GAEZ) database (Fischer et al., 2021). We use Theme 3 rain-fed agro-climatic potential yields under standardized low- and high-input management regimes. These simulations characterize production potential rather than realized county yields or observed crop choices and provide a common measure of agro-climatic production opportunities (Bustos et al., 2016). For each crop, we measure the proportional difference in potential yield between the standardized high- and low-input regimes and then difference this measure across corn and soybeans:

$$RIPA_{ca} = \left(\log Y_{ca}^{C,H} - \log Y_{ca}^{C,L}\right) - \left(\log Y_{ca}^{S,H} - \log Y_{ca}^{S,L}\right),$$

where a indexes the GAEZ reference period. A higher $RIPA$ indicates that the proportional high-versus-low-input potential-yield difference is larger for corn than for soybeans. It measures the relative input-premium margin of the local agro-climatic production environment. We treat this margin as one component of corn–soy comparative advantage, alongside relative yield levels, prices, revenue risk, and other economic conditions.

We combine the GAEZ measures with county-level corn–soy land allocation, pre-planting expected revenues, and revenue-risk measures. Because GAEZ potential yields are observed for only three historical reference periods, we use changes in long-run climate normals to construct annual measures of the crop-specific input premiums and $RIPA$. The analysis focuses on reallocation within counties where corn and soybeans are already an important part of local cropland.

The reduced-form evidence shows a clear difference between persistent spatial production opportunities and shorter-run economic incentives. Counties with a larger relative input premium for corn tend to allocate a larger share of corn–soy acreage to corn, whereas within-county changes in $RIPA$ are weakly related to annual acreage movements. Changes in expected relative revenues are more closely associated with year-to-year reallocation. These results are consistent with $RIPA$ capturing a persistent component of corn–soy comparative advantage rather than short-run variation in profitability.

We incorporate these margins in a discrete-choice land-allocation model in which alternative corn shares depend on expected portfolio revenue, $RIPA$, revenue risk, and distance from the previous year's allocation. Expected revenue and relative input-premium advantage favor higher corn shares, while risk and distance from prior allocation reduce the value of an alternative. The estimated lag-distance penalties imply strong conditional state dependence and remain large after allowing for broad spatial differences in crop allocation and under a predetermined county sample. A one-interquartile-range increase in corn's expected revenue advantage (about \$108 per acre) raises the model-implied corn share by 0.69 percentage points under the estimated persistence penalties. The response is much larger when those penalties are removed.

We also examine the directly observed change in $RIPA$ between the 1990 and 2010 GAEZ anchor years. The changes are spatially heterogeneous, with corn-favoring shifts concentrated in parts of the northern and western production region and modest soybean-favoring shifts elsewhere. The aggregate land-allocation response is small, but model-implied spatial reallocation is considerably larger when the estimated persistence penalties are removed. Because expected revenues, revenue risk, and lagged allocation are held fixed, this exercise isolates the historical relative-production- opportunity channel rather than the total effect of climate change on crop allocation.

The paper makes three contributions. First, we separate changes in climate-conditioned relative production opportunities from persistence in observed land allocation. The estimates show substantial conditional state dependence, creating a wedge between changes in production incentives and short-run acreage responses. This wedge implies that small short-run land-use responses need not imply small longer-run reallocation following persistent changes in comparative advantage. We interpret the estimated persistence broadly rather than as a direct measure of physical adjustment costs.

Second, we use GAEZ agro-climatic potential yields to construct a crop-relative measure of production opportunity (Bustos et al., 2016; Costinot et al., 2016; Fischer et al., 2021). $RIPA$ compares the high-versus-low-input potential-yield premium for corn with the corresponding premium for soybeans. This construction matches the comparative nature of the allocation decision. $RIPA$ measures the relative input-premium margin, not overall corn–soy relative productivity.

Third, we relate land allocation to expected portfolio revenue, relative production opportunity, revenue risk, and prior allocation within a common choice framework (e.g., Fezzi and Bateman, 2011; Hendricks et al., 2014). The model allows changes in crop-allocation incentives to be distinguished from state dependence in prior allocation and quantifies how the latter dampens model-implied acreage responses. The analysis is limited to reallocation within an established corn–soy production system and does not model entry into or exit from that system.

2 Data and Variable Construction

Our data combine county-level land use, weather and climate, GAEZ agro-climatic potential yields, commodity prices, and revenue-risk measures.

2.1 Sample Definition and Land Use

The study area is the 22-state Extended Corn Belt shown in Figure 1, including the central Corn Belt and counties along its western, northern, and southern margins.

We obtain county-level harvested corn and soybean acreage from USDA NASS Quick Stats. Let A^C_{ct} and A^S_{ct} denote harvested corn and soybean acreage in county c and year t. Total cropland acreage, A^{crop}_{ct}, is taken from the Census of Agriculture. Because cropland acreage is observed only in Census years, we assign annual values using a nearest-Census-year step function. We use this measure only to define the corn–soy production sample.

The main land-use outcome is the corn share of combined corn–soybean harvested acreage, q_{ct}. We also define CS_{ct} as the share of total cropland in corn and soybeans:

$$q_{ct} = \frac{A^C_{ct}}{A^C_{ct} + A^S_{ct}}. \tag{1}$$

$$CS_{ct} = \frac{A^C_{ct} + A^S_{ct}}{A^{crop}_{ct}}.$$

The baseline sample retains county-years with $CS_{ct} \geq 0.60$, so corn and soybeans account for at least 60% of local cropland. The resulting panel contains 20,400 county-year observations from 681 counties over 1960–2022.

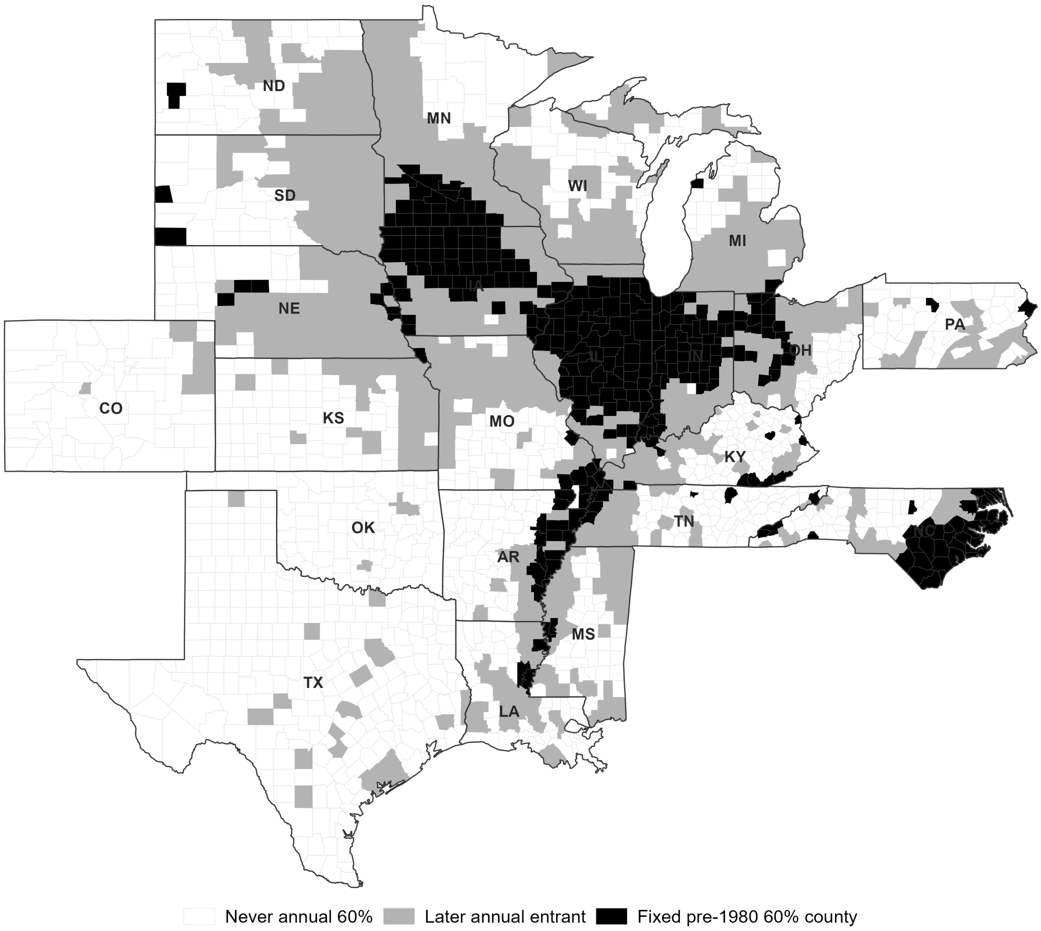

Figure 1: Extended Corn Belt and Analysis Sample

Notes: The Extended Corn Belt includes Arkansas, Colorado, Illinois, Indiana, Iowa, Kansas, Kentucky, Louisiana, Michigan, Minnesota, Mississippi, Missouri, Nebraska, North Carolina, North Dakota, Ohio, Oklahoma, Pennsylvania, South Dakota, Tennessee, Texas, and Wisconsin. Black counties satisfy the predetermined-support criterion based on 1960–1979 land use, requiring corn and soybean harvested acreage to account for at least 60% of total cropland and at least 10 valid pre-period observations. Light-gray counties do not satisfy the predetermined criterion but meet the annual $CS_{ct} \geq 0.60$ threshold in at least one year during 1980–2022. The baseline sample includes county-years that meet this annual threshold, so light-gray counties enter only in qualifying years. White counties never meet the annual threshold during 1980–2022.

Because the baseline threshold is applied contemporaneously, crop composition can affect sample membership. The baseline analysis thus describes reallocation within counties where corn and soybeans are an important part of local cropland, rather than entry into or exit from corn–soy production.

For robustness, we define a predetermined county support using only 1960–1979 land use:

$$\overline{CS}_c^{pre} = \frac{\sum_{t=1960}^{1979} \left(A_{ct}^C + A_{ct}^S\right)}{\sum_{t=1960}^{1979} A_{ct}^{crop}}.$$

A county is included if $\overline{CS}_c^{pre} \geq 0.60$ and at least 10 valid pre-period observations are available. County membership is then held fixed over 1980–2022.

2.2 Weather and Climate Data

We obtain daily maximum temperature, minimum temperature, and precipitation from the PRISM Climate Group's 4-km gridded product and aggregate observations to counties using cropland-area weights. For daily variable x_{gdt},

$$x_{cdt} = \sum_{g \in c} w_{gc} x_{gdt}, \qquad w_{gc} = \frac{A_{gc}^w}{\sum_{h \in c} A_{hc}^w},$$

where A_{gc}^w denotes the cropland area of grid cell g within county c.

Annual weather measures cover the March 1–September 30 growing season. Growing degree days (GDD) measure cumulative thermal exposure between 10°C and 30°C (50°F–86°F), while extreme-heat degree days (ODD) measure exposure above 32°C (90°F). Growing-season precipitation is the sum of daily precipitation over the same period.

For each county-year, we construct trailing 30-year averages of growing-season GDD, ODD, and precipitation. These climate normals enter the expected-yield model and the annual construction of the GAEZ measures. For the GAEZ reference periods ending in 1990, 2000, and 2010, we use the corresponding 1961–1990, 1971–2000, and 1981–2010 climate windows.

2.3 GAEZ-Based Agro-Climatic Potential-Yield Measures

We use Theme 3, *Agro-climatic Potential Yield*, from the FAO Global Agro-Ecological Zones (GAEZ v4) database (Fischer et al., 2021). The data report rain-fed maize and soybean potential yields under standardized low- and high-input management regimes. We aggregate the GAEZ raster data to U.S. counties using the zonal mean of grid cells intersecting each county, with the same procedure applied across crops, input regimes, and reference periods.

The low-input regime represents traditional production with limited external inputs, while the high-input regime represents advanced management using improved varieties, mechanization, nutrient applications, and crop protection. Both regimes are rain-fed, so the comparison does not reflect differences in irrigation. Appendix A provides additional details on the management assumptions.

The historical simulations correspond to the climatic reference periods 1961–1990, 1971–2000, and 1981–2010. Let $a \in \{1990, 2000, 2010\}$ denote the ending year of each period, and let $Y_{ca}^{k,H}$ and $Y_{ca}^{k,L}$ denote county c's high- and low-input potential yields for crop $k \in \{C, S\}$. We define the crop-specific Input Premium Advantage as

$$IPA_{ca}^{k} = \log Y_{ca}^{k,H} - \log Y_{ca}^{k,L}, \tag{2}$$

and the Relative Input Premium Advantage as

$$RIPA_{ca} = IPA_{ca}^{C} - IPA_{ca}^{S}. \tag{3}$$

IPA^k is the proportional difference in potential yield between the standardized high- and low-input regimes for crop k. $RIPA$ compares this difference across corn and soybeans. A higher $RIPA$ indicates a larger high-versus-low-input potential-yield difference for corn relative to soybeans. We treat this relative input-premium margin as one agro-climatic component of corn–soy comparative advantage, not as a complete measure of relative productivity.

Appendix B.1 considers alternative measures of this production margin. A symmetric measure of the high-versus-low-input potential-yield difference is also positively associated with corn allocation, while a measure based only on relative high-input potential yields is negatively associated with corn allocation. The positive $RIPA$ relationship is therefore specific to the relative input-premium margin rather than relative potential-yield levels.

Annual Construction

GAEZ measures are available for only three multi-decadal reference periods, whereas the land-allocation panel is annual. We construct annual series using changes in 30-year climate normals. For $G \in \{IPA^C, IPA^S, RIPA\}$, let $b(c)$ denote county c's earliest usable GAEZ reference period. We estimate

$$G_{ca} - G_{c,b(c)} = \boldsymbol{\gamma}_G' \left(Z_{ca} - Z_{c,b(c)} \right) + u_{ca}^{G}$$

separately for each G by ordinary least squares, where Z contains trailing 30-year growing-season GDD, ODD, and precipitation. Annual values are constructed as

$$\widehat{G}_{ct} = G_{c,b(c)} + \widehat{\gamma}'_G \left(Z_{ct} - Z_{c,b(c)}\right).$$

The construction preserves each county's baseline GAEZ value and uses changes in climate normals to generate annual variation.

The reference period ending in 1990 provides the baseline for nearly all counties. Values before 1990 are backward projections, values after 2010 are forward projections, and values between reference periods are model generated except at the observed endpoints. We therefore treat the annual series as climate-conditioned projections of the GAEZ input-premium measures rather than observed annual GAEZ potential yields.

We assess the sensitivity of the $RIPA$ estimates to this construction. Restricting the land-allocation model to 1990–2010 eliminates the pre-1990 and post-2010 extrapolation periods, while leaving the coefficient on $q_j RIPA_{ct}$ positive and similar to the full-sample estimate. The relationship is also similar under a quadratic climate-normal specification and linear interpolation between GAEZ reference periods. Appendix B.2 and Appendix B.3 report these results.

The historical counterfactual does not use the annual projection. Instead, it uses the observed change in $RIPA$ between the GAEZ reference periods ending in 1990 and 2010:

$$\Delta RIPA_c^{hist} = RIPA_c^{10} - RIPA_c^{90}.$$

Descriptive Validation

We relate the GAEZ measures to observed crop yields and corn–soy allocation. The regressions include year fixed effects, trailing climate-normal controls, and county-clustered standard errors.

Table 1 shows that corn IPA is positively associated with realized log corn yield, while soybean IPA is negatively associated with realized soybean yield. The negative soybean coefficient does not imply that higher-input management reduces soybean productivity. IPA^k measures the proportional difference between standardized high- and low-input potential yields, not the causal effect of input use.

$RIPA$ is positively associated with the corn share of combined corn–soybean acreage. Counties with a larger relative input premium for corn tend to allocate a larger share of corn–

soy acreage to corn. These associations are descriptive, and $RIPA$ captures only one component of corn–soy comparative advantage.

Table 1: Descriptive Validation of GAEZ-Based IPA and RIPA

	Log corn yield	Log soybean yield	Corn share
IPA^C	0.2573***	–	–
	(0.0239)		
IPA^S	–	-1.2678***	–
		(0.1533)	
$RIPA$	–	–	0.2076***
			(0.0227)
Observations	20,400	20,400	20,400
R^2	0.108	0.059	0.173
Year FE	Yes	Yes	Yes
Climate-normal controls	Yes	Yes	Yes

Notes: Standard errors clustered by county are reported in parentheses. The dependent variable in column (3) is the corn share of corn-plus-soybean acreage. All regressions include year fixed effects and trailing climate-normal controls for growing-season GDD, ODD, and precipitation. The estimates are descriptive validation exercises and should not be interpreted as causal effects. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

2.4 Commodity Prices, Expected Returns, and Revenue Risk

We construct expected crop revenue from February futures prices and predicted county yields. Revenue risk is based on deviations of realized revenue from these expectations. Appendix C provides additional construction details.

Pre-Planting Futures Prices. Let P_{tk}^{fut} denote the pre-planting futures price for crop $k \in \{C, S\}$ in year t. We use Chicago Board of Trade December corn and November soybean futures contracts. Expected price is the average daily settlement price over all February trading days, expressed in dollars per bushel.

Expected Yields and Revenue. Let Y_{ctk} denote realized NASS county yield for crop $k \in \{C, S\}$. Expected yield, $\widehat{Y}_{ctk}$, is obtained from the crop-specific log-yield model

$$\log Y_{ctk} = \alpha_{ck} + \delta_k IPA_{ct}^k + \gamma_k q_{c,t-1} + \boldsymbol{\lambda}_k' Z_{ct} + \sum_s \mathbf{1}(c \in s) \left(\eta_{1s}^k t + \eta_{2s}^k t^2 \right) + \nu_{ctk}, \tag{4}$$

where α_{ck} is a crop-specific county fixed effect, IPA_{ct}^k is the crop-specific GAEZ input premium, $q_{c,t-1}$ is the lagged corn share, and

$$Z_{ct} = \left(GDD_{ct}^{30}, ODD_{ct}^{30}, P_{ct}^{30} \right)'$$

contains trailing 30-year climate normals. The model also includes state-specific quadratic time trends and excludes planting-year growing-season weather realizations and year fixed effects. Predicted yields are

$$\widehat{Y}_{ctk} = \exp\left(\widehat{\log Y}_{ctk}\right).$$

Expected crop revenue per acre is

$$\mu_{ctk} = P_{tk}^{fut}\widehat{Y}_{ctk}, \tag{5}$$

and expected portfolio revenue for candidate allocation j is

$$\mu_{ctj} = q_j\mu_{ctC} + (1-q_j)\mu_{ctS}. \tag{6}$$

The reduced-form regressions use expected relative revenue, $\mu_{ctC} - \mu_{ctS}$. The land-allocation model uses expected portfolio revenue evaluated at each candidate corn share. Appendix C.1 describes the expected-yield and revenue construction in more detail.

Revenue Innovations and Portfolio Variance. Realized crop revenue per acre is

$$Rev_{ctk} = P_{s(c),tk}^{NASS}Y_{ctk}, \tag{7}$$

where $P_{s(c),tk}^{NASS}$ is the NASS state-level marketing-year-average price for crop k in the state containing county c. The crop-specific revenue innovation is

$$\varepsilon_{ctk} = Rev_{ctk} - \mu_{ctk}. \tag{8}$$

These innovations include both yield and price forecast errors.

For each year t, crop-specific revenue variances and the corn–soy covariance are calculated from the preceding 15 revenue innovations, $\tau = t-15, \ldots, t-1$:

$$\Sigma_{ct} = \begin{pmatrix} \sigma_{ctC}^2 & \sigma_{ctCS} \\ \sigma_{ctCS} & \sigma_{ctS}^2 \end{pmatrix}.$$

Portfolio revenue variance for candidate allocation j is

$$Var_{ctj} = q_j^2\sigma_{ctC}^2 + (1-q_j)^2\sigma_{ctS}^2 + 2q_j(1-q_j)\sigma_{ctCS}. \tag{9}$$

Expected Shortfall and Marginal Risk Measures. For candidate allocation j, portfolio revenue innovations over the preceding 15 years are

$$\varepsilon_{c\tau j} = q_j\varepsilon_{c\tau C} + (1-q_j)\varepsilon_{c\tau S}.$$

Let $Q_{0.20,ctj}$ denote the empirical 20th percentile of these innovations. Expected Shortfall is

$$ES_{ctj} = -E\left[\varepsilon_{c\tau j} \mid \varepsilon_{c\tau j} \leq Q_{0.20,ctj}\right], \qquad \tau = t-15, \ldots, t-1. \tag{10}$$

The negative sign expresses adverse revenue innovations as positive losses, so larger values of ES_{ctj} indicate greater downside risk. Appendix C.2 gives further details on the portfolio-risk measures.

The reduced-form regressions use marginal risk evaluated around the previous year's corn share. Marginal portfolio variance is

$$MV_{ct} = \frac{2q_{c,t-1}\sigma^2_{ctC} - 2(1-q_{c,t-1})\sigma^2_{ctS} + 2(1-2q_{c,t-1})\sigma_{ctCS}}{10{,}000}. \tag{11}$$

Marginal downside risk is approximated by a symmetric finite difference in Expected Shortfall:

$$MD_{ct} = \frac{1}{100}\frac{ES_{ct}(q_{c,t-1}+h) - ES_{ct}(q_{c,t-1}-h)}{2h}, \qquad h = 0.05. \tag{12}$$

Positive values of MV_{ct} or MD_{ct} indicate that a higher corn share raises the corresponding measure of revenue risk. The denominators are the scalings used in estimation.

The marginal measures enter only the reduced-form regressions. The land-allocation model instead evaluates Var_{ctj} and ES_{ctj} at each candidate allocation. Appendix C.3 describes the marginal risk construction, including the treatment of allocations near the boundaries.

3 Conceptual Framework

We model county-level corn–soy allocation as a function of expected relative returns, relative production opportunities, revenue risk, and prior allocation. Let $q_{ct} \in [0,1]$ denote the share of county c's corn–soy acreage allocated to corn in year t. Conditional on the previous year's allocation, the per-period value of corn share q is

$$\mathcal{V}_{ct}(q) = \Pi_{ct}(q; \Delta r_{ct}, S_{ct}) - \mathcal{R}_{ct}(q; \Omega_{ct}) - \mathcal{C}_{ct}(q, q_{c,t-1}), \tag{13}$$

where Δr_{ct} is expected corn revenue relative to soybean revenue, S_{ct} summarizes relative production opportunity, and Ω_{ct} describes the joint distribution of corn and soybean revenues.

Higher expected corn revenue raises the return to allocating land to corn. Relative production opportunity captures how local conditions favor corn relative to soybeans. We assume

$$\frac{\partial^2 \Pi_{ct}}{\partial q\,\partial S} > 0,$$

so that an improvement in corn's relative production opportunity raises the marginal payoff to corn acreage. The risk term depends on the joint distribution of corn and soybean revenues, so its effect on the preferred corn share need not have a fixed sign.

To represent persistence, let

$$\mathcal{C}_{ct} = \frac{\phi}{2}\left(q - q_{c,t-1}\right)^2, \qquad \phi > 0. \tag{14}$$

A larger ϕ makes larger departures from the previous year's corn share less attractive. This term is a simple representation of persistence rather than a literal adjustment-cost equation. In the empirical model, state dependence may also reflect rotations, machinery and management requirements, persistent local conditions, omitted incentives, or other sources of persistence.

For an interior optimum,

$$\frac{\partial \Pi_{ct}}{\partial q} - \frac{\partial \mathcal{R}_{ct}}{\partial q} - \phi\left(q^*_{ct} - q_{c,t-1}\right) = 0. \tag{15}$$

The second-order condition is

$$D_{ct} \equiv \frac{\partial^2 \mathcal{V}_{ct}}{\partial q^2} = \frac{\partial^2 \Pi_{ct}}{\partial q^2} - \frac{\partial^2 \mathcal{R}_{ct}}{\partial q^2} - \phi < 0.$$

Hence,

$$\frac{\partial q^*_{ct}}{\partial S_{ct}} = -\frac{\partial^2 \Pi_{ct}/\partial q\, \partial S}{D_{ct}} > 0. \tag{16}$$

An improvement in corn's relative production opportunity raises the preferred corn share. The same logic applies to expected relative returns when higher expected corn revenue raises the marginal return to corn acreage. A larger ϕ makes D_{ct} more negative and reduces the magnitude of these responses.

Climate can affect land allocation through relative production opportunities, expected crop revenues, and revenue risk. Persistence can limit the acreage response to changes in any of these margins, so short-run allocation may move little even when relative production incentives change substantially.

In the empirical analysis, expected crop revenues measure the return channel, while the GAEZ-based $RIPA$ measure captures the relative agro-climatic input-premium margin. Portfolio variance and expected shortfall measure revenue risk. State dependence is modeled more flexibly using linear and quadratic distance from prior allocation rather than the single parameter ϕ in Equation (14). We compare model-implied responses under the estimated lag-distance penalties with a benchmark that sets those penalties to zero. The historical GAEZ exercise changes $RIPA$ while holding expected revenues, risk, and prior allocation fixed, so it isolates the relative input-premium channel rather than the full effect of climate change on corn–soy allocation.

4 Reduced-Form Evidence

We examine how corn–soy allocation varies with relative production opportunity, expected relative returns, revenue risk, and prior allocation. The outcome is the corn share defined in Equation (1). The estimates are conditional associations and are not interpreted as causal effects.

4.1 Relative Input-Premium Advantage

We first examine the relationship between corn–soy allocation and $RIPA_{ct}$, distinguishing persistent cross-county differences from within-county changes over time.

Table 2: Relative Suitability and Corn Allocation

	County Means (1)	Panel (2)	Two-Way FE (3)
$RIPA$	0.0717 (0.0608)	0.1779*** (0.0211)	-0.0018 (0.0232)
Observations	681	20,400	20,400
R^2	0.674	0.178	0.859
State FE	Yes	No	No
Year FE	No	Yes	Yes
County FE	No	No	Yes

Notes: The dependent variable is the corn share of corn-plus-soybean acreage. Column (1) collapses the annual panel to county means and includes state fixed effects; standard errors are clustered by state. Column (2) uses the annual county panel and includes year fixed effects. Column (3) additionally includes county fixed effects, so identification comes from within-county changes in $RIPA$ over time. Standard errors in columns (2) and (3) are clustered by county. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 2 shows that the relationship between $RIPA$ and corn allocation is primarily cross-sectional. The county-average specification yields a positive but imprecise coefficient. In the annual panel with year fixed effects, the coefficient is 0.1779 and precisely estimated. Adding county fixed effects reduces the coefficient to -0.0018, which is statistically indistinguishable from zero.

Counties with a larger relative input premium for corn tend to allocate a larger share of corn–soy acreage to corn, but within-county changes in $RIPA$ are not systematically associated with annual acreage movements. This pattern is consistent with $RIPA$ representing a persistent component of relative production opportunity rather than short-run profitability. As discussed in Appendix B.1, the positive allocation relationship also appears for a symmetric measure of the relative input premium but not for a measure based only on relative high-input potential yields.

4.2 Persistence and State Dependence

Corn–soy allocation is highly persistent. Figure 2 summarizes year-to-year changes in the corn share. The mean absolute annual change is 0.030 and the median is 0.022. In 82.5% of transitions, the corn share changes by less than five percentage points.

Panel B groups the corn share into ten 10-percentage-point intervals. About 71.0% of county-year observations remain in the same interval from one year to the next, 28.2% move by one interval, and fewer than 1% move by two or more intervals. These transitions document persistence but do not identify its source.

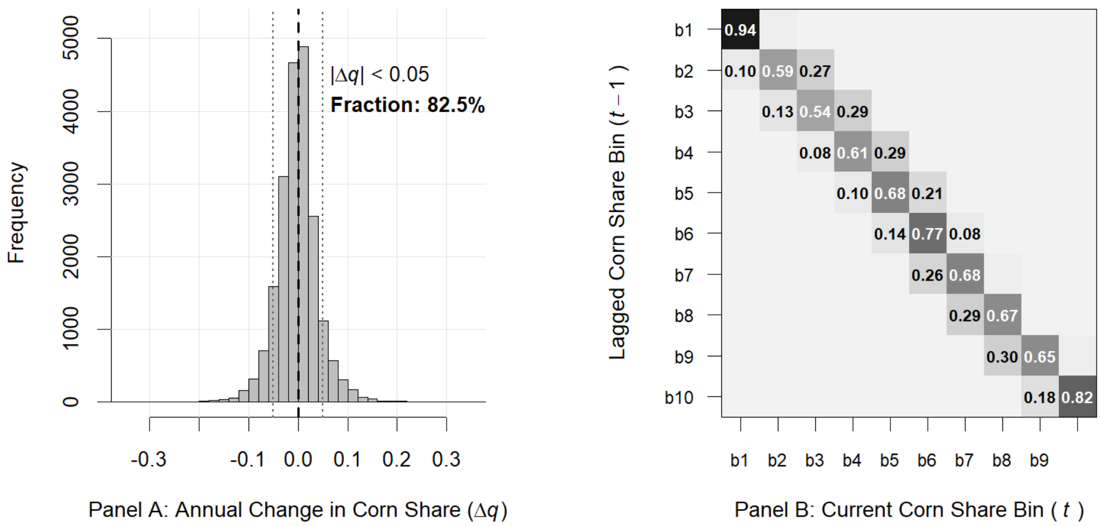

Figure 2: Persistence in Corn–Soy Land Allocation

Notes: Panel A plots the distribution of annual changes in the corn share, $\Delta q_{ct} = q_{ct} - q_{c,t-1}$, for consecutive county-year observations. Panel B presents the empirical transition matrix after grouping the corn share into ten 10-percentage-point intervals represented by $q_j \in \{0.05, 0.15, \ldots, 0.95\}$. Each cell reports the conditional probability of moving from the previous year's interval to the current interval.

We next examine whether persistence remains after conditioning on expected relative returns, $RIPA$, and revenue risk:

$$q_{ct} = \rho q_{c,t-1} + \beta_R \Delta \widehat{R}_{ct} + \beta_S RIPA_{ct} + \beta_V MV_{ct} + \beta_D MD_{ct} + \alpha_c + \lambda_t + \varepsilon_{ct}. \tag{17}$$

Table 3: Persistence and Short-Run Adjustment in Corn Allocation

	Dynamic FE	First Differences	
	(1)	(2)	(3)
Lagged corn share, $q_{c,t-1}$	0.7800***		
	(0.0100)		
Expected relative revenue	0.1430***	0.0692***	0.0694***
	(0.0183)	(0.0119)	(0.0119)
$RIPA$	0.0004	-0.0032	-0.0031
	(0.0053)	(0.0097)	(0.0097)
Marginal variance	0.0017		0.0013
	(0.0015)		(0.0017)
Marginal downside risk	-0.00004		0.00004
	(0.0001)		(0.0001)
Observations	20,400	19,719	19,719
R^2	0.948	0.370	0.370
County FE	Yes	No	No
Year FE	Yes	Yes	Yes

Notes: Column (1) estimates Equation (17) for the level of the corn share and includes county and year fixed effects. Columns (2) and (3) estimate Equation (18) for annual changes in the corn share and include year fixed effects. In Column (1), expected relative revenue, $RIPA$, marginal variance, and marginal downside risk enter in levels. In Columns (2) and (3), they enter as annual changes. Column (3) adds changes in the two risk measures to the specification in Column (2). Standard errors clustered by county are reported in parentheses. *, **, and *** denote statistical significance at the 10, 5, and 1 percent levels, respectively.

Column (1) of Table 3 reports a coefficient of 0.7800 on the lagged corn share. Prior allocation remains strongly associated with current allocation after conditioning on expected relative revenue, $RIPA$, revenue risk, and county and year fixed effects. This state dependence may reflect crop rotations, machinery and management requirements, persistent local conditions, omitted incentives, or other sources of persistence. We do not interpret the lag coefficient as a direct measure of physical adjustment costs.

Expected relative revenue is also positively associated with corn allocation. Its coefficient is 0.1430 when expected relative revenue is measured in thousands of dollars per acre. Higher expected corn revenue relative to soybean revenue is therefore associated with a higher corn share, conditional on prior allocation and the other covariates.

Columns (2) and (3) focus on annual changes:

$$\Delta q_{ct} = \lambda_t + \gamma_R \Delta\left(\Delta \widehat{R}_{ct}\right) + \gamma_S \Delta RIPA_{ct} + \gamma_V \Delta MV_{ct} + \gamma_D \Delta MD_{ct} + u_{ct}. \tag{18}$$

Changes in corn's expected revenue advantage are positively associated with changes in the corn share, with a coefficient of 0.0694 in the full specification. The coefficients on changes in $RIPA$, marginal variance, and marginal downside risk are small and statistically imprecise.

The reduced-form evidence separates two margins that enter the land-allocation model. $RIPA$ is associated mainly with persistent cross-county differences in corn–soy allocation, while expected relative revenues are more closely associated with annual acreage changes. Prior allocation remains strongly predictive after conditioning on both channels and on revenue risk. The model in the next section allows these production, return, risk, and persistence margins to enter the value of each candidate allocation jointly.

5 Empirical Land-Allocation Model

The reduced-form evidence points to four features of corn–soy land allocation: persistent spatial differences in relative agro-climatic production opportunities, time-varying expected returns, revenue risk, and dependence on previous allocation. We combine these components in a discrete-choice land-allocation model. The value of alternative corn shares depends on expected portfolio revenue, $RIPA$, portfolio risk, and distance from the previous allocation. The lag-distance terms capture reduced-form state dependence rather than physical adjustment costs.

5.1 Choice Environment

We represent county-level corn–soy allocation using the same ten 10-percentage-point intervals shown in Figure 2:

$$Q = \{0.05, 0.15, 0.25, 0.35, 0.45, 0.55, 0.65, 0.75, 0.85, 0.95\}. \tag{19}$$

The midpoint q_j represents allocation state j, and the observed choice is the interval containing the realized corn share. This discrete representation approximates an underlying continuous county-level allocation and is not intended to describe the choice set faced by individual producers.

Alternative-specific attributes are evaluated at q_j, while distance from the previous allocation is measured using the continuous lagged corn share $q_{c,t-1}$. This preserves information on the county's prior allocation and avoids mechanically generating state dependence from the choice-grid discretization. We also consider alternative choice grids as a robustness check.

The model is estimated on the baseline corn–soy sample defined in Section 2.1. The analysis therefore concerns reallocation within an established corn–soy production system, not entry into or exit from that system.

5.2 Choice-Specific Payoffs

Let U_{ctj} denote the latent choice-value index for allocation state j:

$$U_{ctj} = V_{ctj} + \xi_{ctj}, \tag{20}$$

where

$$\begin{aligned} V_{ctj} = \alpha_j + \beta_\mu \widetilde{\mu}_{ctj} + \beta_V \widetilde{Var}_{ctj} + \beta_D \widetilde{ES}_{ctj} + \beta_S q_j RIPA_{ct} \\ + \theta_L \left| q_j - q_{c,t-1} \right| + \theta_Q \left(q_j - q_{c,t-1} \right)^2 . \end{aligned} \tag{21}$$

The variables μ_{ctj}, Var_{ctj}, and ES_{ctj} are constructed as described in Section 2.4. For estimation, expected revenue is expressed in thousands of dollars per acre, portfolio variance is divided by 10,000, and expected shortfall by 100. The tildes denote these scaled variables. The alternative-specific constant, α_j, is normalized to zero.

Expected revenue. Expected portfolio revenue varies across candidate allocations with the expected revenue advantage of corn relative to soybeans. Because the soybean revenue component common to all alternatives cancels from the conditional-logit probabilities, a positive β_μ implies that a larger expected corn revenue advantage increases the relative value of higher corn-share states.

Relative input-premium advantage. Relative input-premium advantage enters as $\beta_S q_j RIPA_{ct}$. Because $RIPA_{ct}$ does not vary across alternatives within a county-year, a stand-alone $RIPA$ term would cancel from the conditional-logit probabilities. Interacting it with q_j allows $RIPA$ to affect the relative value of different corn-share states. A positive β_S indicates that higher $RIPA$ is associated with higher corn shares. As defined in Section 2.3, $RIPA$ measures one agro-climatic component of corn–soy comparative advantage.

Revenue risk. The model uses the candidate-specific portfolio variance and expected shortfall measures defined in Section 2.4. Both are evaluated at each q_j, so the risk associated with alternative corn–soy allocations can differ across choices. The reduced-form regressions instead use marginal risk evaluated around the previous allocation. Because portfolio risk depends on the variances and covariance of corn and soybean revenues, variance and expected shortfall need not change monotonically with the corn share.

Persistence and lag-distance penalties. We capture state dependence through the distance between candidate allocation q_j and the previous year's continuous corn share:

$$D^L_{ctj} = |q_j - q_{c,t-1}|,$$
$$D^Q_{ctj} = (q_j - q_{c,t-1})^2.$$

These terms enter Equation (21) with coefficients θ_L and θ_Q. Negative coefficients imply that allocations farther from the previous corn share have lower estimated choice values, conditional on the other observed attributes. The quadratic term allows the penalty to vary with the size of the change.

We interpret θ_L and θ_Q as reduced-form state-dependence terms. Adjustment frictions associated with rotations, machinery, management, input procurement, and marketing may contribute to this persistence, but the coefficients may also capture persistent local conditions, omitted economic incentives, or other sources of state dependence. They are therefore not interpreted as direct estimates of the adjustment-cost parameter ϕ in Section 3.

5.3 Estimation and Inference

We estimate Equation (21) using conditional logit, with each county-year treated as a choice among the ten allocation states. If ξ_{ctj} follows an i.i.d. Type I extreme-value distribution, the probability of choosing state j is

$$P_{ctj} = \frac{\exp(V_{ctj})}{\sum_{\ell \in Q} \exp(V_{ct\ell})}. \tag{22}$$

The parameters are estimated by maximum likelihood, with the interval containing the observed corn share designated as the chosen state.

Identification comes from differences in attributes across candidate allocations within a county-year and from variation in those differences across county-years. Expected revenue varies across alternatives through the interaction between the candidate corn share and the expected corn–soy revenue differential. $RIPA$ enters through $q_j RIPA_{ct}$, while portfolio variance and expected shortfall vary nonlinearly across candidate shares. The lag-distance coefficients are identified from differences between each candidate share and the county's previous allocation. Attributes common to all alternatives within a county-year cancel from the choice probabilities.

The alternative-specific constants α_j capture persistent differences in the relative attractiveness of positions on the allocation grid that are common across counties. As a robustness check, we replace α_j with state-by-allocation fixed effects,

$$\alpha_j \quad \longrightarrow \quad \delta_{s(c),j}.$$

This allows the allocation profile to differ flexibly across states. If the lag-distance coefficients remain large, they are unlikely to reflect only state-level differences in crop composition, although finer spatial heterogeneity and other sources of state dependence may still matter.

The coefficients are interpreted as conditional associations within the choice model, not as causal effects. The $RIPA$ coefficient captures how the relative agro-climatic input premium is associated with higher corn-share states, conditional on expected revenue, portfolio risk, and prior allocation. The lag-distance coefficients measure conditional state dependence and are not interpreted as physical adjustment costs.

The Type I extreme-value assumption implies the usual independence of irrelevant alternatives property. Because the alternatives are adjacent intervals used to approximate a continuous allocation decision, we treat the conditional logit as a parsimonious empirical approximation rather than a literal model of individual producer choice. Results using alternative allocation grids provide a check on this discretization.

Standard errors are clustered by county. Counterfactual confidence intervals are based on a county-level block bootstrap that resamples counties and preserves each county's time series.

6 Empirical Estimates and Model Implications

We estimate the land-allocation model using 20,400 county-year choice situations from 1960 to 2022. Each choice set contains ten candidate corn-share states, yielding 204,000 alternative-level observations.

6.1 Estimation Results

Table 4 reports four specifications that sequentially add relative production opportunity, revenue risk, and lag-distance terms. Because conditional-logit coefficients are estimated on a latent index whose scale can change across specifications, we do not compare coefficient magnitudes mechanically across columns.

Table 4: Discrete-Choice Land-Allocation Model: Incremental Estimates

	(1) Revenue	(2) + Suitability	(3) + Risk	(4) + Adjustment
Expected Revenue ($\widetilde{\mu}_{ctj}$)	2.7870***	1.8012**	3.3199***	27.8725***
	(0.8568)	(0.8885)	(0.8773)	(1.2049)
Relative Suitability ($q_j \times RIPA_{ct}$)		10.1186***	10.1186***	4.2951***
		(1.2564)	(1.2718)	(0.6842)
Portfolio Variance ($\widetilde{Var}_{ctj}$)			-1.0902**	-5.0341***
			(0.4250)	(0.6011)
Downside Expected Shortfall ($\widetilde{ES}_{ctj}$)			-2.0461***	-0.5867
			(0.4586)	(0.5706)
Linear Lag Distance (D^L_{ctj})				-16.2978***
				(1.4697)
Quadratic Lag Distance (D^Q_{ctj})				-73.3442***
				(11.5417)
Alternative-Specific Constants	Yes	Yes	Yes	Yes
Choice Situations	20,400	20,400	20,400	20,400
Alternative-Level Observations	204,000	204,000	204,000	204,000
Log-Likelihood	-33,079.6	-31,806.6	-31,672.4	-13,850.8
AIC	66,179.2	63,635.2	63,370.8	27,731.6

Notes: The dependent variable is the observed corn-share allocation alternative. Expected portfolio revenue is scaled by \$1,000 per acre, portfolio variance by 10,000, and downside expected shortfall by 100. $D^L_{ctj} = |q_j - q_{c,t-1}|$ and $D^Q_{ctj} = (q_j - q_{c,t-1})^2$ measure adjustment distance from the continuous lagged corn share. All specifications include alternative-specific constants, with the first alternative normalized to zero. Standard errors clustered by county are reported in parentheses. *, **, and *** denote statistical significance at the 10, 5, and 1 percent levels, respectively.

Expected returns, $RIPA$, and revenue risk. Column (1) includes expected portfolio revenue and alternative-specific constants. The expected-revenue coefficient is positive and statistically significant. Column (2) adds $q_j RIPA_{ct}$, which also enters positively and is precisely estimated. Conditional on expected revenue, counties with a larger relative input premium for corn assign higher choice values to allocations with larger corn shares.

Column (3) adds portfolio variance and expected shortfall. Both coefficients are negative and statistically significant. Candidate allocations with greater revenue variance or downside exposure therefore have lower estimated choice values, conditional on expected revenue and $RIPA$.

Persistence and lag-distance penalties. Column (4), the preferred specification, adds linear and quadratic distance from the previous continuous corn share. Both coefficients are negative and precisely estimated:

$$\widehat{\theta}_L = -16.2978, \qquad \widehat{\theta}_Q = -73.3442.$$

Allocations farther from the previous year's corn share therefore have lower estimated choice values, conditional on expected revenue, $RIPA$, and risk.

The lag-distance terms also account for a large share of model fit. The log likelihood improves from −31,672.4 in Column (3) to −13,850.8 in Column (4). Previous allocation thus contains substantial information about current corn–soy allocation beyond the observed return, production-opportunity, and risk measures.

We interpret the distance coefficients as reduced-form measures of state dependence. Adjustment frictions associated with rotations, machinery, management, input procurement, or marketing may contribute to this persistence, but the coefficients may also reflect persistent local conditions, omitted incentives, or other sources of state dependence rather than physical adjustment costs.

The other main relationships are similar in the preferred specification. Expected portfolio revenue and $q_j RIPA_{ct}$ enter positively, while portfolio variance enters negatively. Expected shortfall is also negative but not statistically significant. The estimated coefficients are 27.8725 for expected revenue, 4.2951 for $q_j RIPA_{ct}$, −5.0341 for portfolio variance, and −0.5867 for expected shortfall.

Economic magnitude. The coefficients are most naturally interpreted through relative choice probabilities. Consider two candidate states that differ by 10 percentage points in corn share. A \$100-per-acre increase in corn's expected revenue relative to soybeans changes expected portfolio revenue between these states by \$10 per acre, or 0.01 in the scaled revenue measure. With $\widehat{\beta}_\mu = 27.8725$, the log odds of the higher-corn-share state increase by

$$27.8725 \times 0.01 = 0.279.$$

A one-interquartile-range increase in $RIPA$ of approximately 0.262 changes the $RIPA$ component of the log odds between states differing by 10 percentage points by

$$4.2951 \times 0.10 \times 0.262 \approx 0.113.$$

These calculations describe shifts in relative choice values rather than acreage responses. The counterfactual analysis below translates the estimated choice relationships into changes in model-implied corn shares.

6.2 Model Fit and Robustness

For each county-year, the model-implied expected corn share is

$$\widehat{q}_{ct} = \sum_{j=1}^{10} q_j \widehat{P}_{ctj}. \tag{23}$$

Figure 3 compares the model-implied and observed corn shares. The correlation is 0.953, the RMSE is 0.040, and the mean absolute error is 0.029. The model also reproduces the concentration of transitions around prior allocation: 71.3% of model-implied transitions remain in the same allocation interval, compared with 71.0% in the data.

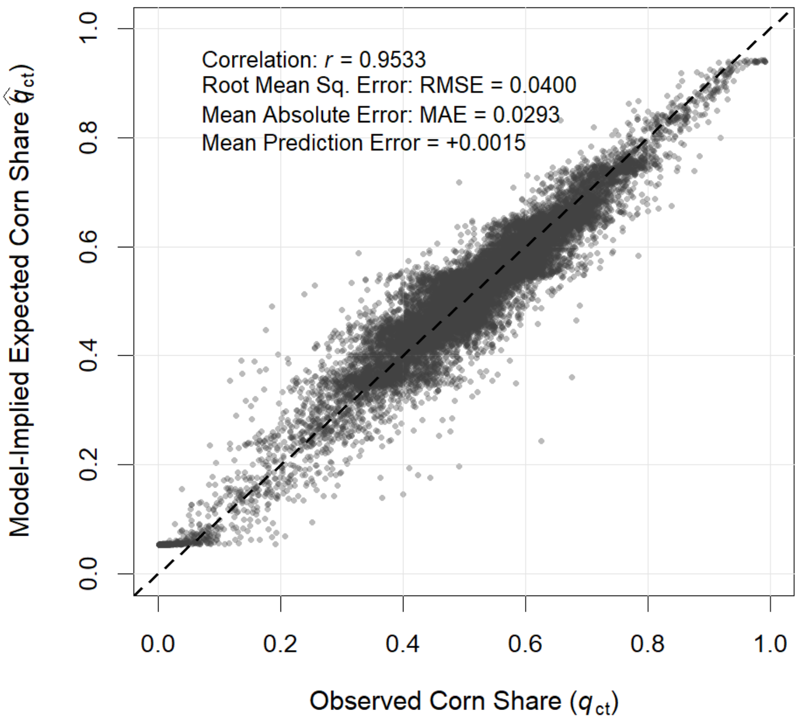

Figure 3: Observed and Model-Implied Corn Shares

Notes: The figure compares the observed continuous corn share q_{ct} with the model-implied expected share defined in Equation (23). The dashed line is the 45-degree line.

The main coefficient signs are unchanged when the corn-share choice set is represented by five, ten, or twenty states. Expected portfolio revenue and $q_j RIPA_{ct}$ remain positive, while portfolio variance and the lag-distance terms remain negative. Expected shortfall is less stable across grids.

We consider alternative measures of relative production opportunity. When corn and soybean IPA enter separately, the corn-IPA coefficient is positive and the soybean-IPA coefficient is negative. A Wald test rejects the equal-and-opposite restriction implied by the one-index $RIPA$ specification ($p = 0.001$). The symmetric relative input-premium measure, $RSIPA$, is positively associated with corn share, while relative high-input potential yield is negatively associated with corn share. These results support interpreting $RIPA$ as a parsimonious measure of the relative input-premium margin, not as a complete measure of corn–soy comparative advantage. Appendix D.1 reports these specifications.

To assess whether the lag-distance estimates are driven by persistent spatial differences in crop allocation, we replace the common allocation-state constants with state-by-allocation fixed effects. Both distance coefficients remain close to their baseline estimates, and the log-likelihood gain from adding the lag-distance terms remains large. The persistence estimates remain substantial after accounting for broad state-level differences in crop composition. Appendix D.2 reports the full estimates.

As a sample check, we re-estimate the model on a predetermined county support defined from 1960–1979 land use and held fixed over 1980–2022. Expected portfolio revenue and $RIPA$ remain positive, portfolio variance remains negative, and the linear lag-distance coefficient remains large and statistically significant. The quadratic distance term is smaller and imprecisely estimated. Across alternative predetermined-support thresholds, the return, $RIPA$, and linear lag-distance estimates are similar, while the quadratic term is less stable. Appendix D.3 reports the full results.

Appendix D.4 considers alternative revenue-risk windows and expected-yield specifications. Expected portfolio revenue and $q_j RIPA_{ct}$ remain positive, while portfolio variance and the lag-distance terms remain negative across specifications.

7 Counterfactual Land Allocation and Revenue-Equivalent Value

We use the estimated land-allocation model to examine changes in expected relative returns, $RIPA$, revenue risk, and estimated state dependence. In each exercise, we change the relevant model attributes, hold the estimated coefficients fixed, and recompute the probabilities of the ten allocation states. We report the resulting change in the expected corn share and the change in allocation value expressed in revenue-equivalent dollars per acre.

These exercises are model-implied comparative statics, not causal estimates of policy, climate, or technology shocks. The standardized exercises vary one component of the choice environment at a time. For the historical exercise, we apply each county's observed change in $RIPA$ between the GAEZ reference periods ending in 1990 and 2010. Expected revenues, revenue risk, and prior allocation are held fixed, so the exercise isolates the allocation response to the historical shift in the relative input-premium margin rather than the total effect of climate change.

7.1 Counterfactual Framework

Let $\widehat{V}^0_{ctj}$ denote the fitted choice-value index under the baseline environment and $\widehat{V}^r_{ctj}$ the corresponding index under counterfactual r. The inclusive value under r is

$$IV^r_{ct} = \log\left[\sum_{j=1}^{10} \exp\left(\widehat{V}^r_{ctj}\right)\right].$$

Let $\widehat{P}^r_{ctj}$ denote the predicted probability of allocation state j under counterfactual r. The model-implied corn share is

$$\widehat{q}^r_{ct} = \sum_{j=1}^{10} q_j \widehat{P}^r_{ctj},$$

with $\widehat{q}^0_{ct}$ defined analogously under the baseline. The change in corn share is

$$\Delta\widehat{q}^r_{ct} = \widehat{q}^r_{ct} - \widehat{q}^0_{ct}.$$

Because expected portfolio revenue enters the choice index in thousands of dollars per acre, we express changes in the inclusive value in expected-revenue units:

$$CV^r_{ct} = \frac{1{,}000}{\widehat{\beta}_\mu}\left(IV^r_{ct} - IV^0_{ct}\right). \tag{24}$$

With $\widehat{\beta}_\mu = 27.8725$, one index unit is equivalent to about \$35.88 per acre in expected revenue. We use CV^r_{ct} as a revenue-equivalent measure of the change in model-implied allocation value, not as a measure of producer surplus or welfare.

Confidence intervals are based on 500 county block-bootstrap replications, with each county's full time series retained when resampled.

7.2 Standardized Counterfactuals

Table 5 reports standardized changes in expected relative returns, $RIPA$, and revenue risk. A one-interquartile-range increase in corn's expected revenue advantage, \$107.58 per acre, raises the model-implied corn share by 0.69 percentage points and the revenue-equivalent allocation value by \$56.18 per acre. A one-IQR decrease produces a similar response in the opposite direction.

Table 5: Main Counterfactual Results

	Corn Share		Revenue-Equivalent Value	
Counterfactual	$\Delta\widehat{q}$ (pp)	95% CI	\$/acre	95% CI
Return advantage: +1 IQR	0.69	[0.64, 0.75]	56.18	[56.09, 56.26]
Return advantage: −1 IQR	-0.70	[-0.75, -0.64]	-55.43	[-55.52, -55.34]
Relative suitability: +1 IQR	0.26	[0.18, 0.36]	20.96	[14.61, 28.86]
Revenue risk: +50% variance	-0.08	[-0.09, -0.06]	-12.70	[-15.38, -10.02]

Notes: $\Delta\widehat{q}$ is the mean change in the model-implied expected corn share, reported in percentage points (pp). Revenue-equivalent value is $CV^r_{ct} = (1{,}000/\widehat{\beta}_\mu)(IV^r_{ct} - IV^0_{ct})$ and is reported in dollars per acre. One IQR of the expected corn revenue advantage is \$107.58 per acre and one IQR of $RIPA$ is 0.2616. The revenue-risk counterfactual increases the variance of the underlying corn and soybean revenue innovations by 50% and recomputes portfolio variance and expected shortfall. All results use the estimated adjustment penalties. The 95% confidence intervals are based on 500 county block bootstrap replications.

Increasing $RIPA$ by one interquartile range, 0.262, raises the corn share by 0.26 percentage points and the revenue-equivalent allocation value by \$20.96 per acre. Expected crop revenues and revenue risk are held fixed, so the response reflects the relative input-premium margin captured by $RIPA$.

For the revenue-risk counterfactual, we multiply corn and soybean revenue innovations by $\sqrt{1.5}$, which increases their variances by 50% while preserving their dependence structure. The model-implied corn share falls by 0.08 percentage points, and the revenue-equivalent allocation value falls by \$12.70 per acre.

7.3 Role of Estimated Persistence

To assess how estimated state dependence affects the counterfactual responses, we multiply the linear and quadratic lag-distance coefficients by $f \in \{1, 0.5, 0\}$. The case $f = 1$ uses the estimated coefficients, $f = 0.5$ halves both coefficients, and $f = 0$ sets them to zero. Baseline and counterfactual probabilities are recomputed under each value of f.

We refer to $f = 0$ as the no-persistence-penalty benchmark. All other estimated choice-value relationships are held fixed. This benchmark does not represent the removal of all physical adjustment costs or other sources of persistence.

Table 6: Persistence and Counterfactual Reallocation

	Estimated Penalties ($f = 1$)	Half-Strength Penalties ($f = 0.5$)	No-Penalty Benchmark ($f = 0$)
Panel A. Expected return advantage: +1 IQR			
Change in corn share (pp)	0.69	1.56	11.09
Revenue-equivalent value ($/acre)	56.18	56.43	57.70
Panel B. Revenue risk: +50% variance			
Change in corn share (pp)	−0.08	−0.17	−0.91
Revenue-equivalent value ($/acre)	−12.70	−12.61	−12.20

Notes: The linear and quadratic lag-distance coefficients are scaled by the value of f shown in each column. Revenue-equivalent value compares the counterfactual and baseline environments under the same value of f. The no-penalty case is a model benchmark rather than an observed policy environment.

For a one-IQR increase in corn's expected revenue advantage, the corn-share response rises from 0.69 percentage points under the estimated penalties to 1.56 percentage points when the penalties are halved and 11.09 percentage points when they are set to zero. The no-persistence-penalty response is about 16 times the response under the estimated coefficients.

The corresponding revenue-equivalent changes are $56.18, $56.43, and $57.70 per acre. Estimated state dependence therefore has a much larger effect on the acreage response than on the revenue-equivalent allocation value.

The revenue-risk counterfactual shows the same pattern. The absolute corn-share response rises from 0.08 percentage points under the estimated penalties to 0.17 with half-strength penalties and 0.91 when the penalties are removed. The revenue-equivalent loss changes only from $12.70 to $12.20 per acre.

The attenuation remains under alternative controls for persistent spatial crop propensity and under a predetermined county support. With state-by-allocation-state fixed effects, the expected-return response is 0.738 percentage points under the estimated penalties and 6.771 percentage points when they are removed. The corresponding $RIPA$ responses are 0.238 and 2.185 percentage points. In the predetermined sample, the expected-return response is 0.401 percentage points under the estimated penalties and 3.512 percentage points without them. Appendix D.2 and Appendix D.3 report the corresponding estimates and counterfactuals.

7.4 Historical GAEZ Changes and Spatial Reallocation

We use the observed county-level change in $RIPA$ between the GAEZ reference periods ending in 1990 and 2010:

$$\Delta RIPA_c^{hist} = RIPA_c^{10} - RIPA_c^{90}. \tag{25}$$

This change is calculated directly from the GAEZ Theme 3 rain-fed agro-climatic potential yields. Changes in long-run temperature and precipitation are shown for context but do not enter the construction of $\Delta RIPA_c^{hist}$.

Across the 681 study counties, mean $\Delta RIPA^{hist}$ is 0.0049, the median is -0.0086, and 39.9% of counties have a positive change. Regional differences are pronounced. Mean $\Delta RIPA^{hist}$ is 0.0597 in the North/West region, compared with -0.0107 in the Core Corn Belt and -0.0160 in the South/East and transition region. The near-zero overall mean therefore reflects offsetting regional changes. Figure 4 shows their spatial distribution, and Appendix E.1 report additional descriptive statistics.

For each county-year, we construct the early-period counterfactual

$$RIPA_{ct}^{E} = RIPA_{ct} - \Delta RIPA_c^{hist}. \tag{26}$$

Expected crop revenues, portfolio variance, expected shortfall, and lagged allocation are held fixed. We then recompute choice probabilities and compare the later and early $RIPA$ environments using the share and revenue-equivalent measures defined above.

Table 7 reports the aggregate counterfactual responses. Panel A uses the primary 2011–2022 evaluation sample. The historical $RIPA$ shift raises the mean model-implied corn share by 0.0079 percentage points under the estimated persistence penalties and by 0.1127 percentage points in the no-persistence-penalty benchmark. The corresponding revenue-equivalent allocation values are \$0.75 and \$0.95 per acre.

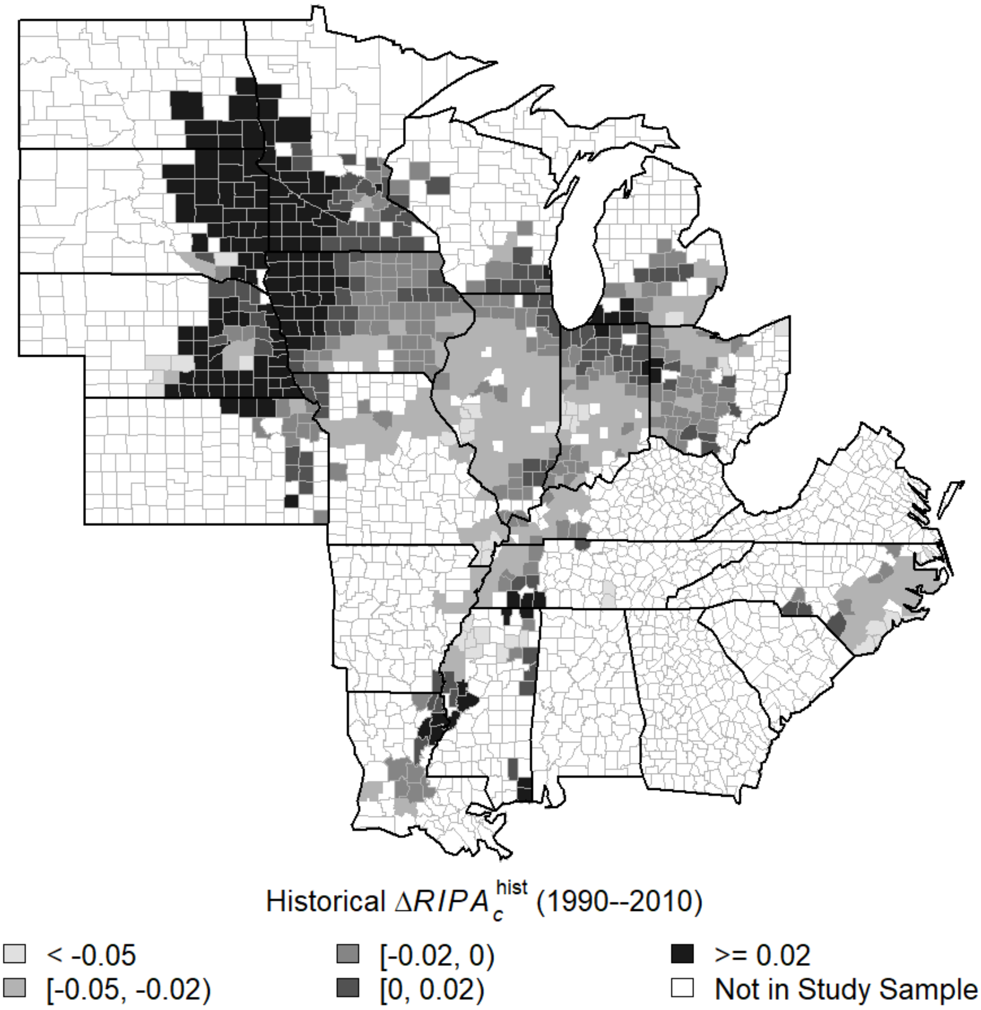

Figure 4: Spatial Distribution of Historical RIPA Shifts, 1990–2010

Notes: The map shows the county-level change in Relative Input Premium Advantage between the GAEZ climatic reference periods ending in 1990 and 2010. Positive values indicate a shift in the relative input-premium measure toward corn; negative values indicate a shift toward soybeans.

Table 7: Historical GAEZ Relative-Production-Opportunity Counterfactual

Specification	$\Delta\widehat{q}$ (pp)	95% CI	Value (\$/ac)	95% CI
Panel A. 2011–2022 ($N = 4{,}829$)				
Estimated penalties ($f = 1$)	0.0079	[0.0026, 0.0135]	0.75	[0.30, 1.21]
No-persistence penalty ($f = 0$)	0.1127	[0.0114, 0.2202]	0.95	[0.34, 1.58]
Panel B. 1960–2022 ($N = 20{,}400$)				
Estimated penalties ($f = 1$)	−0.0001	[−0.0041, 0.0043]	0.12	[−0.25, 0.52]
No-persistence penalty ($f = 0$)	−0.0374	[−0.1406, 0.0754]	0.12	[−0.26, 0.53]

Notes: $\Delta\widehat{q}$ is the change in the model-implied corn share, in percentage points. "Value" is the revenue-equivalent allocation value, $(1{,}000/\widehat{\beta}_\mu)(IV_{ct}^L - IV_{ct}^E)$, associated with the observed change in $RIPA$ between the GAEZ reference periods ending in 1990 and 2010. The estimated specification retains the fitted lag-distance penalties; the no-persistence-penalty benchmark sets both coefficients to zero. Confidence intervals are based on 500 county block-bootstrap replications.

Panel B evaluates the same historical $RIPA$ shift over the full 1960–2022 panel. The mean corn-share responses are -0.0001 percentage points under the estimated penalties and -0.0374 percentage points when the penalties are removed, and neither estimate is statistically distinguishable from zero. The corresponding revenue-equivalent changes are \$0.12 per acre in both specifications. The difference across panels shows that the aggregate mean response is sensitive to the temporal support of the evaluation sample. The predetermined-support historical counterfactual is reported in Appendix E.2.

Table 8 reports the regional and county-level allocation responses to the historical $RIPA$ shift, along with gross corn–soy acreage reallocation.

Table 8: Historical RIPA Shifts and Spatial Reallocation

Panel A. Observed historical $RIPA$ shifts

Region	Mean	Median	Share > 0
Core Corn Belt	−0.0107	−0.0150	25.5%
North/West	0.0597	0.0308	83.1%
South/East and transition	−0.0160	−0.0202	26.9%
All counties	0.0049	−0.0086	39.9%

Panel B. Model-implied corn-share and acreage responses, 2011–2022

Specification	Region	Mean $\Delta\widehat{q}$ (pp)	Mean $\lvert\Delta\widehat{q}\rvert$ (pp)	Gross reallocation (acres/year)
Estimated penalties ($f = 1$)	Core Corn Belt	−0.0116	0.0290	14,645
	North/West	0.0655	0.0781	24,963
	South/East and transition	−0.0168	0.0347	3,420
	All counties	0.0079	0.0431	43,028
No-persistence penalty ($f = 0$)	Core Corn Belt	−0.2292	0.5291	261,340
	North/West	1.1485	1.3798	456,260
	South/East and transition	−0.3557	0.7516	72,434
	All counties	0.1127	0.7956	790,035

Panel C. Distribution of model-implied corn-share responses

Specification	P10	P25	Median	P75	P90
Estimated penalties ($f = 1$)	−0.0481	−0.0301	−0.0086	0.0227	0.0819
No-persistence penalty ($f = 0$)	−1.0732	−0.5346	−0.0852	0.2878	1.2548

Notes: Panel A reports the directly observed county-level change in $RIPA$ between the GAEZ reference periods ending in 1990 and 2010. The Core Corn Belt includes Illinois, Iowa, Indiana, and Ohio; North/West includes Minnesota, North Dakota, South Dakota, Nebraska, and Wisconsin; South/East and transition contains the remaining counties. Panels B and C report model-implied responses over 2011–2022. Gross reallocation is the average annual value of $\sum_c (A^C_{ct} + A^S_{ct})\lvert\Delta\widehat{q}_{ct}\rvert/100$ and therefore measures reallocation between corn and soybeans rather than changes in total cropland. The no-persistence-penalty benchmark sets both lag-distance coefficients to zero while holding the remaining estimated choice-value relationships fixed. The mean absolute corn-share response is 18.46 times larger and gross acreage reallocation is 18.36 times larger under the no-persistence-penalty benchmark.

Under the estimated persistence penalties, the mean corn-share response is 0.0079 percentage points, but the mean absolute response is 0.0431 percentage points. Regional responses differ in sign. The North/West has a mean response of 0.0655 percentage points, compared with −0.0116 in the Core Corn Belt and −0.0168 in the South/East and transition region. The distribution in Panel C shows the same heterogeneity: the median response is slightly negative, while the upper tail is positive.

We measure gross corn–soy acreage reallocation in year t as

$$GR_t = \sum_c \left(A^C_{ct} + A^S_{ct}\right) \frac{|\Delta \widehat{q}_{ct}|}{100}. \tag{27}$$

The acreage base is combined corn and soybean acreage, so GR_t measures reallocation between the two crops rather than changes in total cropland.

Average annual gross reallocation over 2011–2022 is 43,028 acres under the estimated persistence penalties. When the lag-distance coefficients are set to zero, gross reallocation rises to 790,035 acres and the mean absolute corn-share response rises from 0.0431 to 0.7956 percentage points. The distribution of county responses also widens substantially under the no-penalty benchmark.

The historical exercise therefore shows substantial spatial heterogeneity in the allocation response to observed $RIPA$ shifts. The lag-distance terms sharply reduce the magnitude of model-implied reallocation while leaving the regional pattern of responses intact. The comparison measures the role of estimated state dependence and does not identify the effect of removing physical adjustment costs.

8 Conclusion

This paper examines how climate-conditioned production opportunities relate to corn–soy land allocation when production patterns are persistent. We construct a GAEZ-based measure of corn's relative input-premium advantage and combine it with expected crop revenues, revenue risk, and prior allocation in an empirical land-allocation model. $RIPA$ is associated mainly with persistent cross-county differences in crop allocation, while expected relative revenues are more closely related to year-to-year acreage changes.

The estimates indicate substantial conditional state dependence. Allocation states farther from the previous year's corn share have lower estimated choice values, and the lag-distance coefficients remain large after allowing for broad spatial differences in crop allocation. A one-interquartile-range increase in corn's expected revenue advantage raises the model-implied corn share by 0.69 percentage points under the estimated persistence penalties, compared with about 11 percentage points when those penalties are set to zero. This attenuation is similar under alternative spatial controls and on a predetermined county support.

Observed changes in $RIPA$ between the GAEZ reference periods ending in 1990 and 2010 vary substantially across regions. The mean allocation response is small because positive and negative regional responses offset one another, and its sign is sensitive to the geographic support used for evaluation. In the post-2010 sample, average annual gross corn–soy reallocation is about 43,000 acres under the estimated persistence penalties and about 790,000 acres under the no-persistence-penalty benchmark.

The lag-distance coefficients are reduced-form measures of state dependence, not estimates of physical adjustment costs. They may reflect rotations, machinery and management requirements, persistent local production conditions, omitted incentives, or other sources of persistence. The no-persistence-penalty case should therefore be interpreted as a model benchmark rather than a policy experiment.

Persistent production patterns can produce small short-run acreage responses even when relative production incentives change substantially. The historical exercise isolates the relative input-premium channel. It does not capture the full effect of climate change, which may also operate through expected yields and revenues, revenue risk, prices, and land uses outside the corn–soy system.

References

Burke, M. and K. Emerick (2016). Adaptation to climate change: Evidence from US agriculture. *American Economic Journal: Economic Policy 8*(3), 106–140.

Bustos, P., B. Caprettini, and J. Ponticelli (2016). Agricultural productivity and structural transformation: Evidence from Brazil. *American Economic Review 106*(6), 1320–1365.

Costinot, A., D. Donaldson, and C. Smith (2016). Evolving comparative advantage and the impact of climate change in agricultural markets: Evidence from 1.7 million fields around the world. *Journal of Political Economy 124*(1), 205–248.

Cui, X. (2020). Climate change and adaptation in agriculture: Evidence from US cropping patterns. *Journal of Environmental Economics and Management 101*, 102306.

Du, X., D. A. Hennessy, and H. Feng (2014). A natural resource theory of U.S. crop insurance contract choice. *American Journal of Agricultural Economics 96*(1), 232–252.

Fezzi, C. and I. J. Bateman (2011). Structural agricultural land use modelling for spatial agro-environmental policy analysis. *American Journal of Agricultural Economics 93*(4), 1168–1188.

Fischer, G., F. O. Nachtergaele, H. T. van Velthuizen, F. Chiozza, G. Franceschini, M. Henry, D. Muchoney, and S. Tramberend (2021). Global agro-ecological zones v4 – model documentation. Technical report, Food and Agriculture Organization of the United Nations (FAO), Rome.

Gouel, C. (2025). Measuring climate change impacts on agriculture: An equilibrium perspective on supply-side approaches. *Journal of the Association of Environmental and Resource Economists 12*(1), 181–220.

Green, T. R., H. Kipka, O. David, and G. S. McMaster (2018). Where is the USA corn belt, and how is it changing? *Science of the Total Environment 618*, 1613–1618.

Hendricks, N. P., A. Smith, and D. A. Sumner (2014). Crop supply dynamics and the illusion of partial adjustment. *American Journal of Agricultural Economics 96*(5), 1469–1491.

Hornbeck, R. (2012). The enduring impact of the American Dust Bowl: Short- and long-run adjustments to environmental catastrophe. *American Economic Review 102*(4), 1477–1507.

Mendelsohn, R. and W. D. Nordhaus (1994). The impact of global warming on agriculture: A Ricardian analysis. *American Economic Review 89*(4), 1046–1048.

O'Brien, P. L., J. L. Hatfield, C. Dold, E. J. Kistner-Thomas, and K. M. Wacha (2020). Cropping pattern changes diminish agroecosystem services in North and South Dakota, USA. *Agronomy Journal 112*(1), 1–24.

Schlenker, W. and M. J. Roberts (2009). Nonlinear temperature effects indicate severe damages to U.S. crop yields under climate change. *Proceedings of the National Academy of Sciences 106*(37), 15594–15598.

Taylor, C. A. (2026). Irrigation and climate change: Long-run adaptation and its externalities. *Journal of the Association of Environmental and Resource Economists 13*(4), 935–973.

Wright, C. K. and M. C. Wimberly (2013). Recent land use change in the western corn belt threatens grasslands and wetlands. *Proceedings of the National Academy of Sciences 110*(10), 4134–4139.

Appendix

A GAEZ Agro-Climatic Potential Yields and Management Regimes

This appendix provides additional detail on the Global Agro-Ecological Zones (GAEZ v4) measures used to construct the Input Premium Advantage (IPA) and Relative Input Premium Advantage (RIPA), following Fischer et al. (2021). We use Theme 3, *Agro-climatic Potential Yield*, for corn and soybeans under rain-fed conditions. The empirical construction compares potential yields under standardized low- and high-input management regimes. These GAEZ simulations characterize crop production potential under specified agro-climatic and management environments rather than realized county yields or observed technology choices.

GAEZ Agro-Climatic Potential-Yield Framework

GAEZ evaluates crop production potential by matching crop-specific physiological and phenological requirements with prevailing climatic conditions under standardized Land Utilization Types (LUTs). The agro-climatic potential-yield calculations account for radiation and temperature conditions, thermal suitability, crop growth-cycle requirements, rain-fed crop-water balances, and crop-specific responses to water stress and other agro-climatic constraints. Crop calendars are selected within feasible growing periods to identify the timing associated with the highest simulated yield for each crop/LUT and location.

The Theme 3 measures used in this paper should be distinguished from GAEZ's subsequent agro-ecological suitability and attainable-yield assessment. In the GAEZ framework, agro-climatic potential yields are generated in the agro-climatic yield assessment, whereas location-specific soil and terrain constraints are evaluated separately in the agro-edaphic assessment and then integrated into the later agro-ecological suitability and attainable-yield outputs. Accordingly, the measures used here characterize *agro-climatic* production potential and do not incorporate the full set of location-specific soil and terrain limitations represented in the later GAEZ assessment.

Our analysis uses rain-fed simulations throughout. Water-supply regime is therefore held fixed when comparing the low- and high-input management regimes. In particular, the high-versus-low comparison does not represent a transition from rain-fed to irrigated production.

Low- and High-Input Management Regimes

GAEZ distinguishes three generic levels of inputs and management: low, intermediate, and high. We use the low- and high-input regimes as standardized endpoints for constructing IPA and RIPA.

The low-input regime corresponds to GAEZ's traditional-management assumption. Production is characterized as largely subsistence oriented and relies on traditional cultivars and labor-intensive production techniques. The regime assumes no application of plant nutrients, no chemical control of pests and diseases, minimum conservation measures, and fallow requirements to maintain soil fertility.

The high-input regime corresponds to GAEZ's advanced-management assumption. Production is primarily market oriented and uses improved or high-yielding varieties, full mechanization where possible, relatively low labor intensity, optimum applications of nutrients, and chemical control of pests, diseases, and weeds.

These regimes differ along several dimensions simultaneously. Their comparison should therefore not be interpreted as the effect of changing a single input, adopting a particular technology, or moving an observed U.S. farm from low- to high-input production. Rather, the two regimes provide standardized benchmark production environments that allow us to measure how the proportional potential-yield difference associated with input and management intensity varies across locations and crops. Technology and management are thus complements to the agro-climatic production environment; IPA and RIPA do not measure observed technology adoption or technological change.

Historical Reference Periods

GAEZ v4 reports historical agro-climatic assessments for three 30-year reference periods: 1961–1990, 1971–2000, and 1981–2010. For compact notation, we let

$$a \in \{1990, 2000, 2010\}$$

denote the ending year of the corresponding reference period. Thus, an observation indexed by $a = 1990$, for example, represents the GAEZ simulation for the 1961–1990 climatic reference period rather than conditions in the single calendar year 1990.

Input Premium Advantage

For crop $k \in \{C, S\}$, let $Y_{ca}^{k,H}$ and $Y_{ca}^{k,L}$ denote the county-level rain-fed GAEZ agro-climatic potential yields used in the analysis under the high- and low-input regimes, respectively, for reference period a. As defined in Section 2.3, the crop-specific Input Premium Advantage is

$$IPA_{ca}^{k} = \log Y_{ca}^{k,H} - \log Y_{ca}^{k,L} = \log\left(\frac{Y_{ca}^{k,H}}{Y_{ca}^{k,L}}\right). \tag{A1}$$

Thus, IPA^k measures the proportional difference in agro-climatic potential yield between the standardized high- and low-input regimes for crop k.

The magnitude of this input premium is distinct from the level of potential yield itself. A location can have a relatively large IPA^k because the high-versus-low-input yield difference is large even if its absolute agro-climatic potential yield is not especially high. Conversely, a location with high potential yields under both regimes can have a relatively small input premium.

In a small number of observations, the high-input potential yield is no greater than the corresponding low-input potential yield. Such cases should not be interpreted as evidence that greater realized input use causally reduces farm productivity. They are outcomes of the standardized GAEZ crop/LUT simulations under alternative input and management assumptions. Because the regimes differ along several management dimensions and are not observed farm-level treatment states, IPA^k is a relative simulated-yield measure rather than a causal return-to-input parameter.

Relative Input Premium Advantage

The empirical land-allocation problem is comparative, so the relevant measure is not the corn input premium alone. We define

$$RIPA_{ca} = IPA_{ca}^{C} - IPA_{ca}^{S}. \tag{A2}$$

A positive $RIPA$ indicates that the proportional high-versus-low-input potential-yield difference is larger for corn than for soybeans, whereas a negative value indicates a larger input premium for soybeans.

An equivalent representation makes the relative nature of the measure explicit:

$$\begin{aligned} RIPA_{ca} &= \left[\log Y_{ca}^{C,H} - \log Y_{ca}^{S,H}\right] - \left[\log Y_{ca}^{C,L} - \log Y_{ca}^{S,L}\right] \\ &= RA_{ca}^{H} - RA_{ca}^{L}, \end{aligned} \tag{A3}$$

where

$$RA_{ca}^{H} = \log Y_{ca}^{C,H} - \log Y_{ca}^{S,H}$$

and

$$RA_{ca}^{L} = \log Y_{ca}^{C,L} - \log Y_{ca}^{S,L}.$$

Thus, $RIPA$ measures how the relative corn–soy potential-yield environment differs between the standardized high- and low-input regimes.

This crop-relative construction is important for the land-allocation interpretation. A large corn input premium need not favor corn acreage if the corresponding soybean input premium is equally large or larger. A higher $RIPA$ instead indicates that the standardized high-versus-low-input potential-yield difference is larger for corn relative to soybeans. We therefore interpret $RIPA$ as the input-premium dimension of relative agro-climatic production opportunity within the corn–soy system.

This interpretation is deliberately narrower than conventional comparative advantage. $RIPA$ does not summarize relative potential-yield levels across the two crops, nor does it incorporate all location-specific soil and terrain constraints, expected output prices, revenue risk, or other economic determinants of crop profitability. It is therefore a parsimonious indicator of one climate-conditioned production margin rather than a complete measure of corn–soy comparative advantage.

The GAEZ measures have the advantage of being generated under standardized production environments rather than from realized county yields. Realized yields reflect actual input choices, management practices, crop allocation, weather realizations, and other economic decisions that may be jointly determined with land use. The GAEZ potential-yield measures instead provide a common agro-climatic benchmark under specified crop/LUT and management assumptions.

At the same time, this standardization determines the scope of the measure. The low- and high-input regimes are stylized production systems, not observed choices by U.S. farmers. The analysis therefore does not interpret the high-versus-low difference as an estimate of returns to fertilizer, mechanization, improved seed, crop protection, or any other individual technology. Nor does $RIPA$ capture the complete agro-ecological suitability of corn relative to soybeans, because the Theme 3 output used here precedes GAEZ's full integration of location-specific soil and terrain constraints.

Accordingly, throughout the empirical analysis we interpret IPA and RIPA as measures of agro-climatic production opportunity under standardized management regimes. Production technology and management remain important complements to these agro-climatic conditions, but IPA and RIPA themselves do not measure realized technology adoption, technological progress, or the causal productivity effect of changing input use.

B Supplementary Analysis of GAEZ IPA and RIPA

This appendix provides additional evidence on the interpretation and robustness of the GAEZ-based Input Premium Advantage (IPA) and Relative Input Premium Advantage (RIPA) measures. We compare $RIPA$ with alternative measures of relative production opportunity and examine sensitivity to extrapolation and alternative annualization procedures.

B.1 Alternative Measures of Relative Production Opportunity

The baseline $RIPA$ measure compares the crop-specific input premiums:

$$\begin{aligned} RIPA_{ca} &= \left[\log Y_{ca}^{C,H} - \log Y_{ca}^{S,H}\right] - \left[\log Y_{ca}^{C,L} - \log Y_{ca}^{S,L}\right] \\ &= RA_{ca}^{H} - RA_{ca}^{L}, \end{aligned} \tag{B1}$$

where

$$RA_{ca}^{H} = \log Y_{ca}^{C,H} - \log Y_{ca}^{S,H}, \qquad RA_{ca}^{L} = \log Y_{ca}^{C,L} - \log Y_{ca}^{S,L}.$$

Thus, $RIPA$ measures how the corn–soy relative potential-yield environment differs between the standardized high- and low-input GAEZ regimes. It captures the relative input-premium margin rather than relative potential-yield levels more generally.

We consider two alternative measures. The first, HRA, uses only high-input potential yields. Let

$$H_{ca}^{k} = \log Y_{ca}^{k,H}, \qquad k \in \{C, S\}.$$

After standardizing H_{ca}^{k} separately by crop, HRA is defined as standardized corn high-input potential minus standardized soybean high-input potential. Higher HRA indicates a higher relative ranking for corn under the high-input GAEZ simulations.

The second alternative uses a symmetric measure of the high- versus low-input potential-yield gap:

$$SIPA_{ca}^{k} = \frac{2\left(Y_{ca}^{k,H} - Y_{ca}^{k,L}\right)}{Y_{ca}^{k,H} + Y_{ca}^{k,L}}, \tag{B2}$$

and

$$RSIPA_{ca} = SIPA_{ca}^{C} - SIPA_{ca}^{S}. \tag{B3}$$

$RSIPA$ retains the input-premium interpretation without using the log high-to-low ratio.

Table B1 compares the corn-share associations for one-IQR changes in the three measures.

Table B1: Alternative Measures of Relative Production Opportunity and Corn Allocation

Relative-suitability measure	1-IQR effect (pp)	S.E. (pp)
Baseline relative input premium ($RIPA$)	5.43***	0.59
Symmetric relative input premium ($RSIPA$)	2.57**	1.10
High-input-only relative suitability (HRA)	−5.25***	0.42

Notes: The dependent variable is the corn share of corn-plus-soybean acreage. Each estimate reports the change in the corn share, in percentage points, associated with a one-interquartile-range increase in the indicated measure. Measures are entered separately using the baseline reduced-form specification with year fixed effects and no county fixed effects. Standard errors are clustered by county. The estimates are descriptive conditional associations. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A one-IQR increase in $RIPA$ is associated with a 5.43-percentage-point higher corn share, while the corresponding estimate for $RSIPA$ is 2.57 percentage points. The high-input-only HRA measure instead has a negative relationship with corn allocation. The positive $RIPA$ relationship therefore pertains to the relative input-premium margin rather than relative high-input potential-yield levels.

B.2 Robustness to Extrapolation

Table B2 re-estimates the land-allocation model over periods that reduce reliance on extrapolated $RIPA$ values.

Table B2: Robustness to Annualization Extrapolation and Sample Period

Sample	Years	County-years	$q_j \times RIPA_{ct}$
Full panel	1960–2022	20,400	4.2951***
			(0.6842)
Post-1990 panel	1990–2022	10,890	4.512***
			(0.648)
Bounded reference-period window	1990–2010	6,930	4.831***
			(0.715)

Notes: Entries report the coefficient on $q_j \times RIPA_{ct}$ from the preferred land-allocation specification with expected revenue, portfolio variance, downside expected shortfall, and adjustment-cost terms included; county-clustered standard errors are in parentheses. The full-panel row corresponds to column (4) of Table 4. The post-1990 sample eliminates pre-1990 backward projection but retains the post-2010 forward-projection years. The 1990–2010 sample is bounded by the observed GAEZ reference periods ending in 1990 and 2010 and therefore eliminates both pre-1990 and post-2010 extrapolation. Annual values between the observed GAEZ reference periods remain generated by the climate-normal annualization procedure. *** $p < 0.01$.

The coefficient on $q_j RIPA_{ct}$ is 4.295 (s.e. 0.684) over 1960–2022, 4.512 (s.e. 0.648) over 1990–2022, and 4.831 (s.e. 0.715) over 1990–2010. The 1990–2022 specification eliminates pre-1990 backward projection, while the 1990–2010 specification excludes both the pre-1990 and post-2010 extrapolation portions of the panel. Annual values between the GAEZ reference periods remain generated by the climate-normal projection. The positive $RIPA$ relationship is similar across the three samples.

B.3 Alternative Annualization Procedures

Table B3 compares the baseline linear climate-normal projection with a quadratic specification and linear interpolation between GAEZ reference periods.

Table B3: Alternative Annualization Procedures for GAEZ IPA and RIPA

	County-anchored R^2		Model fit	$q_j \times RIPA_{ct}$	
Method	Corn IPA	Soy IPA	Log likelihood	Coef.	S.E.
Linear climate-normal projection	0.965	0.948	−13,850.8	4.2951	0.6842
Quadratic climate-normal projection	0.968	0.951	−13,931.10	4.615	0.598
Linear reference-period interpolation	0.712	0.685	−14,022.15	3.112	0.745

Notes: The table compares annualization methods designed to represent slowly evolving GAEZ production opportunity: the baseline linear climate-normal projection, a quadratic climate-normal projection, and a linear reference-period interpolation. County-anchored R^2 evaluates reference-period levels after preserving the county baseline GAEZ value. The final two columns report the coefficient and county-clustered standard error on $q_j \times RIPA_{ct}$ in the corresponding land-allocation specification. The baseline linear climate-normal projection corresponds to the preferred full-panel specification in Table 4.

The coefficient on $q_j RIPA_{ct}$ is 4.295 (s.e. 0.684) under the baseline linear projection, 4.615

(s.e. 0.598) under the quadratic specification, and 3.112 (s.e. 0.745) under linear interpolation. The positive allocation relationship is similar across these alternative slow-moving constructions of the annual $RIPA$ series.

C Expected Revenue and Revenue-Risk Construction

This appendix provides additional details on the expected-revenue and revenue-risk variables defined in Section 2.4. Expected crop revenue combines February futures prices with predicted county yields. Revenue risk is calculated from the preceding 15 years of deviations between realized and expected crop revenue. The reduced-form regressions use marginal risk measures evaluated around the previous year's corn share, while the land-allocation model evaluates risk at each candidate allocation.

C.1 Expected Yield and Revenue Construction

Expected yields are constructed separately for corn and soybeans using the log-yield specification in Equation (4). The model includes county fixed effects, the crop-specific GAEZ input premium, the previous year's corn share, trailing 30-year climate normals, and state-specific quadratic time trends. Planting-year growing-season weather realizations and year fixed effects are excluded.

The climate controls are the trailing 30-year averages of growing degree days, extreme-heat degree days, and growing-season precipitation described in Section 2. County fixed effects absorb persistent differences in yield levels across counties, while the state-specific trends allow yield growth to differ across states. The crop-specific GAEZ input premium allows predicted yields to vary with the climate-conditioned production measure used elsewhere in the analysis.

Predicted yields are obtained by exponentiating the fitted log yields. They are combined with the February futures prices defined in Section 2.4 to construct expected crop revenue per acre. Expected relative revenue in the reduced-form regressions is the difference between expected corn and soybean revenue,

$$\Delta\mu_{ct} = \mu_{ctC} - \mu_{ctS}.$$

The land-allocation model instead forms expected portfolio revenue for each candidate corn share using Equation (6).

C.2 Revenue Innovations and Portfolio Risk

Realized crop revenue combines county yield with the corresponding NASS state-level marketing-year-average price. The revenue innovation is realized revenue less expected revenue, as defined in Equation (8). It therefore reflects both yield and price forecast errors relative to the pre-planting revenue measure.

For each county-year, the corn and soybean revenue variances and their covariance are calculated from the preceding 15 annual revenue innovations. These moments determine portfolio variance at any corn share through Equation (9).

Expected Shortfall measures downside exposure over the same 15-year window. For a given corn share, annual corn and soybean revenue innovations are combined using the corresponding portfolio weights. Expected Shortfall is the negative mean of innovations in the lower 20 percent of this empirical distribution, as defined in Equation (10). The sign convention expresses adverse revenue outcomes as positive losses, so larger values indicate greater downside risk.

Portfolio variance and Expected Shortfall depend on the corn share. The land-allocation model therefore recalculates both measures at each candidate allocation rather than assigning a single county-year risk measure to all choices.

C.3 Marginal Risk Measures in the Reduced-Form Analysis

The reduced-form regressions require a county-year measure of how revenue risk changes with the corn share. We evaluate this change around the previous year's allocation, $q_{c,t-1}$.

Marginal portfolio variance, MV_{ct}, is the derivative of portfolio variance with respect to the corn share, evaluated at $q_{c,t-1}$ and scaled by 10,000 as shown in Equation (11). A positive value indicates that increasing the corn share from the previous allocation raises portfolio variance.

Marginal downside risk, MD_{ct}, is calculated using the finite difference in Equation (12), with $h = 0.05$. Thus, MD_{ct} compares Expected Shortfall at corn shares five percentage points above and below the previous allocation. One-sided differences are used when the symmetric change would fall outside the feasible interval $[0, 1]$. The measure is scaled by 100 for estimation. A positive value indicates that increasing the corn share raises downside exposure.

The scaling of MV_{ct} and MD_{ct} changes coefficient units but not the underlying measures. These marginal variables enter only the reduced-form regressions. The land-allocation model uses portfolio variance and Expected Shortfall evaluated directly at each candidate corn share.

D Additional Model Robustness

This appendix examines alternative measures of relative production opportunity, the sensitivity of the land-allocation estimates to state-specific allocation propensities, predetermined county supports, and alternative revenue-risk and expected-yield specifications.

D.1 Alternative Relative Production-Opportunity Measures

The baseline specification uses

$$RIPA_{ct} = IPA^C_{ct} - IPA^S_{ct},$$

which imposes equal-and-opposite coefficients on the crop-specific input premiums. We relax this restriction by allowing corn and soybean IPA to enter separately:

$$\begin{aligned} V_{ctj} = \alpha_j + \beta_\mu \widetilde{\mu}_{ctj} + \beta_V \widetilde{Var}_{ctj} + \beta_D \widetilde{ES}_{ctj} \\ + \beta_C q_j IPA^C_{ct} + \beta_S q_j IPA^S_{ct} + \theta_L |q_j - q_{c,t-1}| + \theta_Q (q_j - q_{c,t-1})^2. \end{aligned} \tag{D1}$$

We also replace $RIPA$ with the symmetric relative input-premium measure $RSIPA$ and the high-input-only relative potential-yield measure HRA as defined in B.1.

Table D1 reports the alternative specifications. The separate-IPA specification yields a positive corn-IPA interaction and a negative soybean-IPA interaction. A Wald test rejects

$$H_0 : \beta_C + \beta_S = 0$$

with $p = 0.001$, so the equal-and-opposite restriction imposed by $RIPA$ does not hold exactly.

A one-IQR increase in $RIPA$ raises the model-implied corn share by about 0.27 percentage points, while the corresponding response to $RSIPA$ is about 0.22 percentage points. The high-input-only HRA measure produces a response of approximately -0.28 percentage points. The positive baseline relationship therefore pertains to the relative input-premium margin rather than relative high-input potential-yield levels.

Table D1: Alternative Relative-Suitability Specifications in the Land-Allocation Model

Specification or diagnostic	Result
Panel A. One-IQR model-implied corn-share responses	
Baseline relative input premium ($RIPA$)	0.27 pp
Symmetric relative input premium ($RSIPA$)	0.22 pp
High-input-only relative suitability (HRA)	−0.28 pp
Panel B. Unrestricted crop-specific IPA specification	
$q_j \times IPA^C_{ct}$ coefficient	Positive
$q_j \times IPA^S_{ct}$ coefficient	Negative
Wald test: $H_0 : \beta_C + \beta_S = 0$	$p = 0.001$

Notes: Panel A reports the mean change in the model-implied expected corn share, in percentage points, associated with a one-interquartile-range increase in the indicated relative-suitability measure, holding expected revenues, revenue risk, and lagged allocation fixed. $RIPA$ is the baseline Relative Input Premium Advantage. $RSIPA$ uses a symmetric high–low potential-yield gap for each crop before differencing corn and soybeans. HRA compares crop-standardized log high-input potential yields and does not use low-input potential. Panel B reports the signs of the crop-specific suitability interactions from a specification that replaces $q_j RIPA_{ct}$ with $q_j IPA^C_{ct}$ and $q_j IPA^S_{ct}$. The Wald test evaluates the equal-and-opposite restriction imposed by the baseline index. All specifications retain expected portfolio revenue, portfolio variance, expected shortfall, linear and quadratic adjustment distance, and alternative-specific constants.

D.2 State Dependence and Persistent Spatial Crop Propensity

The baseline model uses common allocation-state constants. To allow persistent crop-allocation patterns to differ across states, we replace these constants with state-by-allocation-state fixed effects:

$$V_{ctj} = \delta_{s(c),j} + \beta_\mu \widetilde{\mu}_{ctj} + \beta_V \widetilde{Var}_{ctj} + \beta_D \widetilde{ES}_{ctj} + \beta_S q_j RIPA_{ct} + \theta_L \left| q_j - q_{c,t-1} \right| + \theta_Q \left(q_j - q_{c,t-1} \right)^2 . \tag{D2}$$

The fixed effects $\delta_{s(c),j}$ allow each state to have its own set of allocation-state constants.

Table D2 compares models with common and state-specific allocation effects, with and without the lag-distance terms.

Table D2: Lag-Distance Terms and Persistent Spatial Crop Propensity

	No lag	Lag-distance terms
Panel A. Common allocation effects		
Expected portfolio revenue	3.3199***	27.8725***
	(0.8773)	(1.2049)
$q_j RIPA_{ct}$	10.1186***	4.2951***
	(1.2718)	(0.6842)
Portfolio variance	−1.0902**	−5.0341***
	(0.4250)	(0.6011)
Expected shortfall	−2.0461***	−0.5867
	(0.4586)	(0.5706)
Linear lag distance	–	−16.2978***
		(1.4697)
Quadratic lag distance	–	−73.3442***
		(11.5417)
Log likelihood	−31,672.4	−13,850.8
Log-likelihood gain	–	17,821.6
AIC	63,370.8	27,731.6
Panel B. State × allocation effects		
Expected portfolio revenue	5.3806***	30.6377***
	(1.0807)	(1.2852)
$q_j RIPA_{ct}$	7.4372***	4.0758***
	(1.8666)	(0.9307)
Portfolio variance	1.0980**	−3.3784***
	(0.5244)	(0.6022)
Expected shortfall	−5.1454***	−2.5204***
	(0.6265)	(0.6013)
Linear lag distance	–	−15.1438***
		(1.4774)
Quadratic lag distance	–	−72.3979***
		(11.6149)
Log likelihood	−26,382.2	−13,467.7
Log-likelihood gain	–	12,914.5
AIC	53,096.4	27,253.4
Panel C. Counterfactual responses with state × allocation effects		
	Estimated penalties	No-penalty benchmark
	($f = 1$)	($f = 0$)
Expected relative revenue: +1 IQR	0.738	6.771
$RIPA$: +1 IQR	0.238	2.185
County-year observations	20,400	
Choice states	10	

Notes: Panels A and B report conditional-logit estimates with common allocation-state effects and state-by-allocation-state fixed effects, respectively. The lag specifications include linear and quadratic distance from the previous year's continuous corn share. County-clustered standard errors are in parentheses. Panel C reports changes in the model-implied corn share, in percentage points, using the state-by-allocation specification. The no-penalty benchmark sets both lag-distance coefficients to zero while holding the remaining estimates fixed. Expected portfolio revenue is scaled by $1,000 per acre, portfolio variance by 10,000, and expected shortfall by 100. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

With common allocation effects, adding the lag-distance terms improves the log likelihood by 17,821.6. The corresponding improvement is 12,914.5 with state-by-allocation fixed effects. The linear distance coefficient changes from -16.2978 to -15.1438, and the quadratic coefficient from -73.3442 to -72.3979. Both remain precisely estimated. The persistence estimates therefore remain substantial after allowing broad spatial differences in crop allocation, although finer local heterogeneity and omitted persistent incentives may remain.

Panel C reports the corresponding counterfactual responses. With state-by-allocation fixed effects, a one-IQR increase in expected relative revenue raises the corn share by 0.738 percentage points under the estimated penalties and by 6.771 percentage points under the no-penalty benchmark. The corresponding $RIPA$ responses are 0.238 and 2.185 percentage points. Thus, the attenuation associated with the lag-distance terms remains when persistent allocation patterns are allowed to vary across states.

D.3 Predetermined County Support

The baseline sample applies the 60% corn–soy criterion annually, so contemporaneous crop composition can affect sample membership. Section 2.1 defines a predetermined county support using 1960–1979 land use and retains qualifying counties throughout 1980–2022.

The pre-period rule identifies 322 counties before complete-case requirements are imposed. Of these, 281 counties contribute observations to the fixed-support estimation. Another 722 counties fail the pre-period rule but satisfy the annual 60% criterion at least once during 1980–2022. Their geographic distribution is shown in Figure 1.

Table D3 separates the change in sample definition from the change in estimation period. Column (1) reports the baseline 1960–2022 model, Column (2) applies the annual rule over 1980–2022, and Column (3) uses the predetermined support over 1980–2022.

Expected portfolio revenue and $RIPA$ remain positive in the predetermined sample, while portfolio variance remains negative. The linear lag-distance coefficient remains large and statistically significant. The quadratic term is smaller and imprecisely estimated. The $RIPA$ coefficient is substantially larger in the predetermined sample, although raw conditional-logit coefficients are not directly comparable across samples because the scale of the choice index may differ.

Table D3: Annual and Predetermined Corn–Soy Sample Definitions

	Annual 60%		Predetermined 60%
	1960–2022 (1)	1980–2022 (2)	1980–2022 (3)
Panel A. Land-allocation estimates			
Expected revenue	27.8725***	23.7906***	17.7702***
	(1.2049)	(1.2704)	(1.7759)
$q_j RIPA_{ct}$	4.2951***	4.7614***	28.5883***
	(0.6842)	(0.7130)	(2.4145)
Portfolio variance	−5.0341***	−4.8339***	−3.0696***
	(0.6011)	(0.5843)	(0.8960)
Expected shortfall	−0.5867	0.3220	−0.1066
	(0.5706)	(0.5989)	(0.6166)
Linear lag distance	−16.2978***	−18.2367***	−21.1957***
	(1.4697)	(1.5863)	(2.0537)
Quadratic lag distance	−73.3442***	−57.3600***	−16.3810
	(11.5417)	(12.2003)	(14.5897)
Choice situations	20,400	16,647	10,193
Counties	681	675	281
Mean corn–soy share	0.806	0.820	0.825
Annual 60% rule	Yes	Yes	No
Predetermined support	No	No	Yes
Log likelihood	−13,850.8	−11,374.1	−7,309.0
Panel B. Counterfactual response on predetermined support			
		Estimated penalties ($f = 1$)	No-penalty benchmark ($f = 0$)
Expected relative revenue: +1 IQR	–	0.401	3.512

Notes: Column (1) reproduces the preferred specification. Column (2) restricts the annual 60% sample to 1980–2022. Column (3) uses county support defined from 1960–1979 land use and held fixed over 1980–2022. County-clustered standard errors are in parentheses. Panel B reports changes in the model-implied corn share, in percentage points, for the predetermined sample. The no-penalty benchmark sets both lag-distance coefficients to zero. Expected revenue is scaled by \$1,000 per acre, portfolio variance by 10,000, and expected shortfall by 100. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Panel B reports the corresponding expected-return counterfactual. On the predetermined support, a one-IQR increase in expected relative revenue raises the corn share by 0.401 percentage points under the estimated penalties and by 3.512 percentage points under the no-penalty benchmark. The attenuation associated with the lag-distance terms therefore remains when county support is fixed using pre-period land use.

Alternative Predetermined Thresholds

We construct predetermined county supports using 50%, 60%, and 70% corn–soy shares over 1960–1979, with the same minimum-data requirement. Table D4 reports selected model estimates for each support.

Expected portfolio revenue and $q_j RIPA_{ct}$ remain positive and statistically significant across all three support definitions. The linear lag-distance coefficient is also stable, ranging from -21.012 to -24.912, and is significant at the 1% level in each sample. Portfolio variance remains negative but becomes imprecisely estimated at the 70% threshold. The quadratic lag-distance coefficient is less stable: it is negative at the 50% and 60% thresholds, statistically significant only at the 10% level at 50%, and turns positive but remains insignificant at 70%. The main return, $RIPA$, and linear state-dependence estimates are therefore robust to the predetermined-support threshold, while the quadratic distance term is sensitive to the narrower samples.

Table D4: Sensitivity to Predetermined Corn–Soy Support Thresholds

Threshold	50%	60%	70%
Counties	405	281	117
Observations	14,587	10,193	4,284
Expected portfolio revenue	15.855***	17.770***	25.849***
	(1.382)	(1.776)	(2.958)
$q_j RIPA_{ct}$	20.361***	28.588***	31.045***
	(2.501)	(2.415)	(4.125)
Portfolio variance	−3.522***	−3.070***	−2.122
	(0.721)	(0.896)	(1.535)
Linear lag distance	−21.012***	−21.196***	−24.912***
	(1.439)	(2.054)	(2.554)
Quadratic lag distance	−18.832*	−16.381	12.817
	(10.112)	(14.590)	(17.167)

Notes: Predetermined county support is defined using the ratio of summed corn-plus-soybean harvested acreage to summed total cropland acreage over 1960–1979, with thresholds of 50%, 60%, and 70% and at least 10 valid pre-period observations. Qualifying counties are retained over 1980–2022. Entries report coefficient estimates from the preferred 10-state conditional-logit specification, with county-clustered standard errors in parentheses. Expected portfolio revenue is scaled by $1,000 per acre and portfolio variance by 10,000. Lag distance is measured relative to the continuous previous-year corn share. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

D.4 Alternative Risk and Yield Specifications

Table D5 compares alternative revenue-risk windows, specifications that enter portfolio variance and expected shortfall separately, and an alternative expected-yield specification.

Expected portfolio revenue and $q_j RIPA_{ct}$ remain positive across these specifications, while portfolio variance and the lag-distance coefficients remain negative. When entered separately,

Table D5: Robustness to Revenue-Risk and Expected-Yield Specifications

	10-year risk (1)	20-year risk (2)	Variance only (3)	ES only (4)	Linear trend (5)
Expected portfolio revenue	27.5244***	27.6562***	27.9140***	25.6152***	26.9926***
$q_j \times RIPA$	4.3653***	4.2993***	4.2616***	4.7660***	4.3361***
Portfolio variance	−3.2129***	−5.7586***	−5.4097***	–	−5.1502***
Expected shortfall	−0.7392*	−0.2951	–	−3.7916***	−0.5494
Linear adjustment	−16.2902***	−16.3003***	−16.2911***	−16.3722***	−16.2840***
Quadratic adjustment	−73.3256***	−73.2243***	−73.4136***	−72.4366***	−73.2263***
Choice states	10	10	10	10	10
Alternative-specific constants	Yes	Yes	Yes	Yes	Yes

Notes: The preferred specification in Table 4 uses a trailing 15-year history of corn and soybean revenue innovations and state-specific quadratic trends in the expected-yield equations. Columns (1) and (2) use 10- and 20-year revenue-risk histories. Columns (3) and (4) include portfolio variance and expected shortfall separately. Column (5) replaces state-specific quadratic trends with state-specific linear trends in the expected-yield equations. Expected portfolio revenue is scaled by \$1,000 per acre, portfolio variance by 10,000, and expected shortfall by 100. Adjustment distance is measured from the continuous previous-year corn share. Inference uses standard errors clustered by county; standard errors are omitted here to keep the robustness table compact. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

portfolio variance and expected shortfall are both negative and statistically significant. The alternative expected-yield specification leaves the main coefficient signs unchanged.

E Additional Historical RIPA Results

This appendix provides additional evidence for the historical $RIPA$ exercise in Section 7.4. We report changes in long-run climate conditions and observed $RIPA$, examine the distribution of model-implied allocation responses, and repeat the historical counterfactual on the predetermined county support.

E.1 Historical Climate Conditions and RIPA Shifts

Table E1 reports changes in trailing 30-year growing-season climate normals and the observed county-level change in $RIPA$ between the GAEZ reference periods ending in 1990 and 2010. The climate changes provide descriptive context; the historical $RIPA$ change used in the counterfactual is calculated directly from the GAEZ reference-period data.

Across the 681 study counties, mean growing degree exposure increases by 1.91 degree-days, mean extreme-heat exposure increases by 0.11 degree-days, and mean March–September precipitation changes by −29.47 mm. Mean $\Delta RIPA^{hist}$ is 0.0049, the median is −0.0086, and

Table E1: Historical Changes in Climate Normals and Relative Suitability, 1990–2010

	1990	2010	Change (SD)
Panel A. Climate normals			
Growing-season GDD (10–30°C)	3358.0	3359.9	1.91 (31.02)
Extreme-heat ODD (> 32°C)	47.64	47.75	0.11 (11.81)
Precipitation (mm)	676.2	673.2	−2.95 (50.84)
Panel B. Historical change in relative suitability			
Mean $\Delta RIPA^{hist}$			0.0049
Standard deviation			0.0594
25th percentile			−0.0274
Median			−0.0086
75th percentile			0.0156
Share positive			39.9%

Notes: Panel A reports county-level means of trailing 30-year climate normals in 1990 and 2010 and the mean change across counties, with the standard deviation of the change in parentheses. Climate variables reflect March–September totals. Panel B summarizes the observed historical change in relative agro-ecological suitability, $\Delta RIPA_c^{hist} = RIPA_c^{10} - RIPA_c^{90}$, across the 681 counties in the analysis sample. The suitability change is constructed by differencing the observed 2010 and 1990 GAEZ anchor-year $RIPA$ values.

39.9% of counties have a positive change. The positive mean is therefore driven by relatively large corn-favoring changes in a subset of counties rather than a uniform shift across the study region.

Figure E1 shows the distribution of observed $RIPA$ changes and the corresponding model-implied corn-share responses over 2011–2022. The response distribution is considerably wider when the estimated persistence penalties are removed.

The historical counterfactual holds expected revenues, revenue risk, and lagged allocation fixed. It therefore isolates the relative input-premium channel represented by $RIPA$ rather than the total effect of historical climate change. Confidence intervals in Table 7 are based on 500 county block-bootstrap replications in which counties are resampled with their complete time series and the land-allocation model is re-estimated. The observed county-level GAEZ changes in $RIPA$ are treated as data.

E.2 Predetermined-Support Historical Counterfactual

We repeat the historical $RIPA$ exercise using the predetermined county support defined from 1960–1979 land use. County membership is therefore fixed before the later evaluation period.

For the 2011–2022 fixed-support sample, the historical $RIPA$ shift reduces the model-implied corn share by 0.0840 percentage points under the estimated persistence penalties and

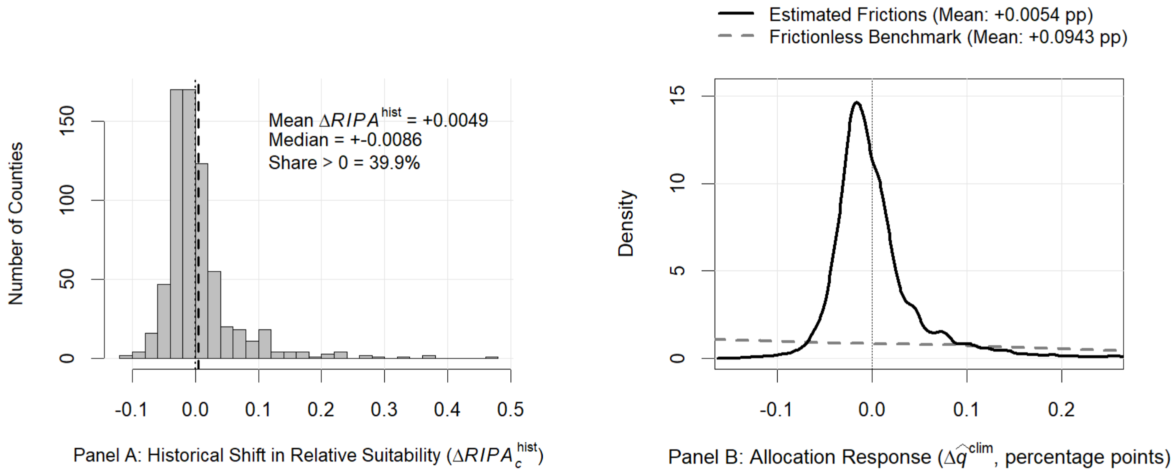

Figure E1: Historical RIPA Shifts and Model-Implied Land-Allocation Responses

Notes: Panel A plots the distribution across the 681 study counties of $\Delta RIPA_c^{hist} = RIPA_c^{10} - RIPA_c^{90}$ between the GAEZ reference periods ending in 1990 and 2010. The dashed line indicates the sample mean (0.0049). Panel B plots kernel density estimates of the model-implied change in the corn share, in percentage points, over the 2011–2022 evaluation sample under the estimated persistence penalties ($f = 1$) and the no-persistence-penalty benchmark ($f = 0$).

Table E2: Historical GAEZ Relative-Suitability Counterfactual on the Predetermined County Support

Evaluation sample	2011–2022	1980–2022
Panel A. Change in expected corn share (percentage points)		
Estimated persistence penalties ($f = 1$)	-0.0840	-0.0804
No-persistence-penalty benchmark ($f = 0$)	-0.6638	-0.7676
Absolute response ratio ($f = 0/f = 1$)	7.90×	9.55×
Panel B. Revenue-equivalent allocation value ($/acre)		
Estimated persistence penalties ($f = 1$)	-4.43	-5.56
No-persistence-penalty benchmark ($f = 0$)	-4.49	-6.13
Panel C. Evaluation-sample composition		
County-year observations	2,426	10,193
Counties	260	281

Notes: The table recalculates the historical relative-suitability counterfactual on the predetermined county support defined using 1960–1979 land-use information. The historical shift is $\Delta RIPA_c^{hist} = RIPA_c^{10} - RIPA_c^{90}$, using observed GAEZ values for the reference periods ending in 1990 and 2010. The $f = 1$ rows use the fixed-support land-allocation estimates with the estimated linear and quadratic lag-distance penalties. The $f = 0$ rows set both lag-distance coefficients to zero in the later- and early-suitability environments while holding the remaining estimated choice-value relationships fixed. The absolute response ratio is the magnitude of the $f = 0$ corn-share response divided by the magnitude of the $f = 1$ response. Panel B reports revenue-equivalent changes in model-implied allocation value in dollars per acre.

by 0.6638 percentage points when the lag-distance coefficients are set to zero. The corresponding revenue-equivalent changes are $-\$4.43$ and $-\$4.49$ per acre.

The aggregate sign differs from the baseline post-2010 exercise. Mean $\Delta RIPA^{hist}$ is -0.0106 within the predetermined support, and 30.0% of counties have a positive change. The difference therefore reflects the geographic composition of the sample and the spatial heterogeneity of the observed $RIPA$ shifts.

The sign of the mean historical response is sensitive to geographic support, but the attenuation from estimated persistence remains. Removing the lag-distance penalties increases the absolute corn-share response by nearly eightfold in the fixed-support sample while changing the revenue-equivalent allocation value only slightly.